

Resource-Adaptive Decentralized LoRA: Failure Analysis and a Quantity-Skew Benchmark

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Abstract

Can heterogeneous clients train useful low-rank adapters through peer communication without sacrificing accuracy? We study this question in two settings. An initial CIFAR-100 experiment with disjoint label domains fails to preserve pooled accuracy: adaptive-rank, domain-weighted LoRA reaches $6.90 \pm 2.03\%$ versus $57.23 \pm 0.59\%$ for pooled training. Controlled diagnostics identify local label allocation and repeated heterogeneous projection as important limitations; synchronized peer gradients reproduce pooled predictions when every client retains full rank. We then evaluate a separately specified quantity-skew benchmark using RoBERTa-base, SST-2, ten ring-connected clients, and an independent reconstruction of published Dec-LoRA. Seven methods are trained at five paired seeds, with unequal dataset sizes, independent capacity assignments, and equal local optimization budgets. Adaptive rank with sample-size weighting reaches $94.24 \pm 0.25\%$ best validation accuracy versus $94.54 \pm 0.24\%$ for rank-16 Dec-LoRA. The paired difference is -0.30 percentage points, with a 95% interval of $[-0.76, 0.16]$; this does not establish equivalence. Serialized training traffic falls by 31.81%, but the feasible rank-4 comparator reaches $94.50 \pm 0.33\%$, while the proposed method uses 8.92% more traffic. Final-round adaptive accuracy is also lower than rank-16 Dec-LoRA. All 36 full-budget runs complete, and independent evaluation reproduces predictions from their 72 best/final checkpoints. The quantity-skew study demonstrates a functioning transformer protocol and a measured tradeoff, but no superiority or established accuracy parity. The original complete-model resource claim remains unsupported, and logical data locality supplies no privacy guarantee.

1 Introduction

LoRA adapts a pretrained model by training low-rank matrices while keeping the original weights fixed [3]. Federated learning permits participants to retain their examples locally [6]; decentralized optimization replaces a dedicated aggregation server with communication between participating peers [1, 4]. Together these ideas suggest a practical objective: match ordinary pooled-data LoRA accuracy while allowing participants with smaller computing budgets to train smaller adapters.

Our original criterion is accuracy preservation relative to pooled training; an improvement over pooled training is not required. A subsequent study asks the narrower question of competitiveness against a published decentralized method on one shared task with unequal client dataset sizes. These are distinct estimands: matching another federated baseline cannot establish pooled parity, and the new study does not include a pooled transformer comparator. The proposed benefit also has two distinct parts: keeping raw examples local is a property of the training workflow, whereas protecting information in exchanged updates requires a separate privacy mechanism and threat model. A possible hospital collaboration motivates the problem but is not an evaluated application.

Combining the components introduces difficulties that isolated experiments do not resolve. LoRA factors are not identifiable, heterogeneous ranks induce different scaling and compression,

and nonuniform contribution weights must survive the gossip normalization. The final model also matters: an adapter assembled centrally by an evaluator is not necessarily a model produced by the stated peer protocol.

We first evaluate AH-LoRA’s original composition of scaled effective-update aggregation, a source-pinned adaptive-rank controller, conservative domain weights, weighted Metropolis gossip, and final peer assembly. The completed pooled comparison is negative. Paired controls examine label allocation and factor refactorization; an oracle-initialized stress test isolates information loss without training; exact gradient synchronization provides a uniform-rank positive control. Training-only holdout studies subsequently test retained-base residuals and partial gradients. These results are retained rather than replaced by a more favorable task.

The new experiment changes the scientific question deliberately: one SST-2 task is divided into unequal, label-stratified shards, and resource-aware variants are compared directly with Dec-LoRA [2]. All seven methods share the same partition, graph, random seed, and local sample stream. The design separates adaptation and sample-size weighting, includes a feasible uniform rank-4 comparator, and measures communication beyond adapter factors. Its 35 paired runs and one equal-size reconstruction anchor complete the registered finite campaign. This is an empirical study of executable protocols, not a new convergence theorem or a validated private-learning system.

2 Related Work and Comparator Scope

Dec-LoRA performs local fine-tuning followed by separate neighbor averages of the two LoRA factors [2]. We select its REALM 2025 RoBERTa-base/SST-2 ring setting because it directly addresses decentralized fine-tuning and supplies a tractable published comparator. No author implementation was verified, so our baseline is an independent reconstruction. We pin the published workshop version rather than combine its settings with changed scores from later arXiv versions. Unspecified optimizer, scaling, target-module, classifier, and split choices are disclosed in the experimental design.

Effective-product consensus followed by low-rank factorization is already central to DeCAF [7]. Our product-space operation is therefore not claimed as a new aggregation principle. The unconstrained product-space arm is a mechanism control on the Dec-LoRA task, not a numerical reproduction of DeCAF’s CLIP or LLaMA experiments. Likewise, weights proportional to dataset size are conventional empirical-risk weights [6], not new domain-allocation theory. The question is whether a particular combination of capability-bounded adaptation, these weights, and graph-constrained merging delivers a useful measured tradeoff. Static quantities yield static sample weights; they must not be conflated with the dynamic domain signals in the historical study.

3 Learning Objective and Evaluation

Let $\mathcal{D}_i$ contain n_i training examples at client i , with $n = \sum_i n_i$. For either model family, the intended sample-weighted objective is $F(\theta) = \sum_i (n_i/n) F_i(\theta)$, where F_i is the client’s mean cross entropy and θ denotes trainable parameters. The two studies use different architectures and endpoints, which we do not pool statistically. In the original CIFAR-100 study, let $\phi(x) \in \mathbb{R}^{512}$ be the shared frozen feature extractor. A client’s logits are

$$z_i(x) = (W_0 + \Delta W_i)\phi(x) + b_0, \quad \Delta W_i = \frac{\alpha}{r_i} B_i A_i, \quad 1 \leq r_i \leq R_i^{\max}. \quad (1)$$

The frozen head (W_0, b_0) , frozen features, and LoRA initialization are shared within each seed. Local loss is the empirical cross entropy

$$f_i(\Delta W) = \frac{1}{n_i} \sum_{(x,y) \in \mathcal{D}_i} \ell((W_0 + \Delta W)\phi(x) + b_0, y). \quad (2)$$

Ordinary pooled LoRA trains one rank- R adapter directly on all examples:

$$\min_{A,B} F_{\text{pool}}(A, B) = \sum_i \frac{n_i}{n} f_i\left(\frac{\alpha}{R} B A\right), \quad R = 16. \quad (3)$$

Sample-weighted federated comparators target the same empirical weighting. Quality and domain arms instead use round-dependent weights $\pi_i^t \propto n_i q_i^t \lambda_i^t$. These weights change the training emphasis; all methods are still evaluated on the same unweighted full test set.

The original study’s primary endpoint is the final assembled rank-16 model’s accuracy on all 10,000 test examples. For seed s and method m , we report the paired difference in percentage points,

$$\Delta_{m,s} = 100 \left(\text{Acc}(\widehat{\Delta W}_{m,s}; \mathcal{T}) - \text{Acc}(\Delta W_{\text{pool},s}; \mathcal{T}) \right). \quad (4)$$

A prespecified noninferiority margin was not available; we report observed differences and sample standard deviations without equating a nonsignificant difference with equivalence. Personalized accuracy, $N^{-1} \sum_i \text{Acc}(\Delta W_i; \mathcal{T}_i)$, is a secondary diagnostic. It evaluates different models on different local test shards and cannot replace the pooled-model comparison.

The SST-2 study instead follows the published best-validation endpoint. If $a_{m,s,t}$ is the assembled model’s percentage accuracy on the 872 labeled validation examples after round t , its primary score and paired effect are

$$b_{m,s} = \max_{1 \leq t \leq 20} a_{m,s,t}, \quad d_s = b_{\text{adaptive-sample},s} - b_{\text{Dec-LoRA16},s}. \quad (5)$$

The first occurrence resolves ties; round-20 accuracy is secondary. Across $S = 5$ paired seeds, we report $\bar{d}$, sample standard deviation s_d , and the descriptive interval $\bar{d} \pm t_{0.975, S-1} s_d / \sqrt{S}$. These unadjusted intervals describe seed variation on the same validation set, not sampling uncertainty on a new test population. No equivalence margin is introduced after seeing results. Repeated validation selection prevents interpreting $b_{m,s}$ as hidden-test performance.

4 Methodology

4.1 Capability-constrained adaptive rank

The integrated experiment uses the Project 1 controller’s decision rule rather than a replacement loss-threshold heuristic. Capability ceilings are $R_i^{\max} \in \{4, 8, 16\}$ and the feasible menu is

$$\mathcal{R}_i = \{r \in \{2, 4, 6, 8, 12, 16, 24, 32\} : r \leq R_i^{\max}\}. \quad (6)$$

A temporary probe at the client’s ceiling measures gradient stable rank before local training. For each nonzero two-dimensional LoRA factor gradient G , define $s(G) = \|G\|_F^2 / \|G\|_2^2$; the client signal s_i^t is the median over these gradients. The probe uses one fixed, seeded local training minibatch, independently of the training shuffle. No test labels enter the controller.

The smoothed demand and candidate target are

$$d_i^t = \text{clip}(s_i^t, 0, R_i^{\max}), \quad \bar{d}_i^t = 0.7\bar{d}_i^{t-1} + 0.3d_i^t, \quad (7)$$

$$r_i^* = \text{nearest}_{\mathcal{R}_i} \left(\max \left\{ 2, 0.5c_i R_i^{\max}, \min(\bar{d}_i^t, R_i^{\max}) \right\} \right), \quad (8)$$

where $c_i \in \{0, 1/2, 1\}$ identifies the capability tier; the first demand observation initializes the moving average. Ties in the candidate quantizer choose the smaller rank, subject to its capability floor. The deployed policy starts at the ceiling, holds it for two warm-up rounds, and imposes a minimum feasible rank of at least $R_i^{\max}/2$. An upward request requires $r_i^* > r_i$ and $\bar{d}_i^t \geq 1.10r_i$; a downward request requires $r_i^* < r_i$ and $\bar{d}_i^t \leq 0.85r_i$. Two consecutive requests in the same direction move one menu step. A post-training quality score, defined below, falling more than 5% below its preceding moving average forces restoration to the ceiling at the next decision. Quality is smoothed with weights 0.8/0.2. Optional residual-demand modulation in the generic controller is disabled here.

A rank increase from r to r' preserves the current update by scaling each existing factor by $\sqrt{r'/r}$. New A rows are randomly initialized and their corresponding B columns are zero, preserving the output while allowing the added directions to learn. A decrease uses the truncated effective-update SVD. Gradient probes and quality evaluations are logged separately from training; a smaller training rank does not by itself establish a reduction in total computation. The controller is source-pinned to Project 1; its one-minibatch probe is an explicit adaptation of Project 1's multi-minibatch measurement procedure to this benchmark.

4.2 Quality and domain contribution weights

After local optimization, client i evaluates cross entropy ℓ_i^t on its fixed training probe and sets

$$q_i^t = (1 + \ell_i^t)^{-1}, \quad p_i^t = \frac{n_i q_i^t}{\sum_j n_j q_j^t}. \quad (9)$$

Both sides of the fixed/adaptive comparison use this same quality definition. Domain weighting uses five Project 2 signals. Let n_{ic} be training class counts, $v_{ic} = n_{ic}/n_i$, $g_c = \sum_i n_{ic}/n$, and let

$$u_i^t = \text{vec}(\Delta W_i^{t,\text{local}} - \Delta W_i^{t,\text{pre}}), \quad \bar{u}^t = N^{-1} \sum_i u_i^t. \quad (10)$$

The feature vector comprises Jensen–Shannon divergence to the pooled training histogram, Euclidean and cosine distance to the mean update, normalized class entropy, and class imbalance:

$$\begin{aligned} x_{i1} &= \text{JS}(v_i, g), & x_{i2} &= \|u_i - \bar{u}\|_2, & x_{i3} &= 1 - \frac{u_i^\top \bar{u}}{\|u_i\|_2 \|\bar{u}\|_2}, \\ x_{i4} &= -\frac{\sum_{c:v_{ic}>0} v_{ic} \log v_{ic}}{\log C}, & x_{i5} &= \frac{\max_c n_{ic}}{\min_{c:n_{ic}>0} n_{ic}} \left(1 + \frac{\#\{c : n_{ic} = 0\}}{C} \right). \end{aligned} \quad (11)$$

Cosine distance is zero when its denominator is at most 10^{-12} . For distributions a, b and $m = (a+b)/2$, $\text{JS}(a, b) = \frac{1}{2} \text{KL}(a\|m) + \frac{1}{2} \text{KL}(b\|m)$ and $\text{KL}(a\|b) = \sum_c a_c \log(a_c/b_c)$. The implementation adds 10^{-12} and renormalizes at each divergence call.

The conservative allocator directly imports the canonical Project 2 function. Each feature is standardized across clients and clipped to $[-3, 3]$, with a constant feature contributing zero. With

$K = 5$, its score and bounded multiplier are

$$\begin{aligned} z_{ik} &= \text{clip}\left(\frac{x_{ik} - \bar{x}_k}{\sigma_k}, -3, 3\right), & a_i &= K^{-1} \sum_k z_{ik}, \\ e_i &= \frac{\exp(\text{clip}(a_i, -4, 4))}{\sum_j p_j \exp(\text{clip}(a_j, -4, 4))}, & h_i &= 1 + 0.10(e_i - 1), \end{aligned} \quad (12)$$

$$\lambda_i = \text{clip}(ch_i, 0.85, 1.15), \quad \sum_i p_i \lambda_i = 1. \quad (13)$$

Bisection determines the positive scalar c . The resulting gossip target is

$$\pi_i = \frac{n_i q_i \lambda_i}{\sum_j n_j q_j \lambda_j}. \quad (14)$$

Quality-only arms set $\lambda_i = 1$; sample-only arms set both $q_i = \lambda_i = 1$. Domain weighting is tested for its effect on the parity objective, without requiring it to improve accuracy over quality-only training.

4.3 Effective-update aggregation and weighted gossip

LoRA factors can be gauge transformed as $(B, A) \mapsto (BQ, Q^{-1}A)$ for any invertible Q . This leaves BA unchanged but generally changes factor averages. AH-LoRA therefore sends the factors and reconstructs $(\alpha/r)BA$ at the receiver. The shared α and source rank are used in every forward pass and merge.

For the undirected training graph, form the symmetric Metropolis proposal

$$Q_{ij} = \frac{1}{1 + \max(\deg(i), \deg(j))} \quad \text{on edges}, \quad Q_{ii} = 1 - \sum_{j \neq i} Q_{ij}. \quad (15)$$

To retain nonuniform contribution weights, the actual kernel is

$$P_{ij}^t = Q_{ij} \min\left(1, \frac{\pi_j^t}{\pi_i^t}\right) \quad (i \neq j), \quad P_{ii}^t = 1 - \sum_{j \neq i} P_{ij}^t. \quad (16)$$

It is row stochastic, preserves graph support, and satisfies detailed balance:

$$\pi_i P_{ij} = Q_{ij} \min(\pi_i, \pi_j) = \pi_j P_{ji}, \quad \pi^\top P = \pi^\top. \quad (17)$$

It is generally neither symmetric nor doubly stochastic. A separate validator checks these weighted identities; the uniform-mean convergence claims of the earlier symmetric runner are not applied to this kernel.

After local training, each client forms and compresses

$$Y_i^t = \sum_j P_{ij}^t \Delta W_j^{t, \text{local}}, \quad \Delta W_i^{t+1} = C_{r_i}^t(Y_i^t). \quad (18)$$

If $Y = U \Sigma V^\top$, the stored factors are

$$B = \frac{U_{:r} \Sigma_{:r}^{1/2}}{\sqrt{\alpha/r}}, \quad A = \frac{\Sigma_{:r}^{1/2} V_{:r}^\top}{\sqrt{\alpha/r}}, \quad C_r(Y) = U_{:r} \Sigma_{:r} V_{:r}^\top. \quad (19)$$

This is the minimum-Frobenius-error rank- r approximation, with discarded energy $\|Y - \mathcal{C}_r(Y)\|_F^2 = \sum_{k>r} \sigma_k(Y)^2$. Before compression, the weighted mean is preserved for the current π . After compression its deviation is exactly $\sum_i \pi_i [\mathcal{C}_{r_i}(Y_i) - Y_i]$. Since local optimization, ranks, and weights change over time, these algebraic identities do not imply convergence or accuracy parity. The optional error-feedback extension stores $m_i^{t+1} = Y_i^t + m_i^t - \mathcal{C}_{r_i}(Y_i^t + m_i^t)$; it is evaluated only in the separate historical ablation.

4.4 Peer exchange of weighting information

The domain policy requires global training statistics. These inputs are explicitly delivered over the graph rather than supplied through an uncharged evaluator callback. Each client provides (n_i, q_i) ; domain arms additionally provide its 100-bin training histogram and 51,200-element effective-update change. A breadth-first spanning tree gathers these records to a participating root peer and broadcasts the gathered records over the same tree. Each peer independently computes Equations (11)–(14) from its received view, and equality of the resulting weights is checked.

This straightforward allgather is intentionally fully accounted for. If record i has L_i numeric fields and the gather subtree at peer v is S_v , its gather cost is $8 \sum_{i \in S_v} (L_i + 1)$ bytes, including one source identifier per record. Each broadcast edge carries $8 \sum_i (L_i + 1)$ bytes. Thus

$$B_{\text{control}}^t = 8 \sum_{v \neq \text{root}} \sum_{i \in S_v} (L_i + 1) + 8(N - 1) \sum_i (L_i + 1). \quad (20)$$

The numerical transport model uses eight-byte scalars and identifiers. The simulator shares read-only array references internally but charges full logical transmissions. This exchange reveals all contributed records to every peer; it is not secure aggregation.

4.5 Final assembly and resource accounting

A final model is produced by the peer protocol itself. Before testing, we choose the first client with the largest capability as deployment root and use a spanning tree of the training graph. Let $\text{ch}(i)$ denote tree children. Each peer accumulates a dense weighted numerator and scalar mass,

$$H_i = \pi_i \Delta W_i + \sum_{j \in \text{ch}(i)} H_j, \quad M_i = \pi_i + \sum_{j \in \text{ch}(i)} M_j. \quad (21)$$

Every non-root sends (H_i, M_i) to its parent. Only the root refactorizes:

$$\widehat{\Delta W} = \mathcal{C}_{16}(H_{\text{root}}/M_{\text{root}}). \quad (22)$$

There is no intermediate truncation in this reduction. It constructs the weighted mean before the final rank projection using $N - 1$ existing graph edges. It requires dense temporary storage and a root capable of the final SVD. The root is a participating peer, not an external training server; failure tolerance and root rotation are not evaluated.

For a source adapter with rank r_j , the training payload is $S_j = r_j(512 + 100)$ fp32 scalars. Let $E_t = \{(i, j) : i \neq j, P_{ij}^t > 0\}$. The deployment payload accounting is

$$B_{\text{total}} = 4 \sum_{t=1}^T \sum_{(i,j) \in E_t} S_j^t + \sum_{t=1}^T B_{\text{control}}^t + (N - 1)(4 \cdot 51200 + 8). \quad (23)$$

The last term is one final peer assembly. Earlier assemblies used to draw evaluation curves are recorded separately and do not alter training. FedAvg already returns its centrally aggregated

model and does not require the peer reduction; pooled training has no inter-client payload. Static sample-count exchange, graph setup, packet framing, and subsequent model dissemination are excluded. We report training rank-sample products, $\sum_{t,i} r_i^t n_i$, and probe work separately; this proxy excludes SVD, feature extraction, optimizer, and communication costs and is not a measured total-FLOP or hardware-speedup claim.

4.6 Quantity-weighted transformer adapters

The quantity-skew study applies LoRA to query and value projections in all 12 RoBERTa-base blocks [5]. For adapted layer ℓ at peer i , write

$$X_{i\ell} = \frac{\alpha}{r_i} B_{i\ell} A_{i\ell}, \quad h'_\ell = (W_{0\ell} + X_{i\ell}) h_\ell + b_{0\ell}, \quad \alpha = 16. \quad (24)$$

Each effective adapter update has shape 768×768 ; the pretrained backbone and projection bias $b_{0\ell}$ are frozen, but the complete RoBERTa classification head is trained. Denote its dense projection and output-layer weights and biases collectively by ϑ_i . Its intervening tanh and dropout operations have no trainable parameters. Unlike the historical cached-feature experiment, gradients pass through the transformer’s adapted layers.

With ten peers on a ring, the proposal has $Q_{ii} = Q_{i,i-1} = Q_{i,i+1} = 1/3$ (indices modulo ten). Uniform arms use $\pi_i = 1/10$; sample-weighted arms use $\pi_i = n_i/n$. Equation (16) then defines the graph-supported kernel. Counts and weights remain fixed throughout a run. For common rank r , the reconstructed Dec-LoRA operation is

$$A_{i\ell}^+ = \sum_j Q_{ij} A_{j\ell}^{\text{local}}, \quad B_{i\ell}^+ = \sum_j Q_{ij} B_{j\ell}^{\text{local}}, \quad \vartheta_i^+ = \sum_j Q_{ij} \vartheta_j^{\text{local}}. \quad (25)$$

The effective-product controls instead use

$$Y_{i\ell} = \sum_j P_{ij} X_{j\ell}^{\text{local}}, \quad X_{i\ell}^+ = C_{r_i}(Y_{i\ell}), \quad \vartheta_i^+ = \sum_j P_{ij} \vartheta_j^{\text{local}}. \quad (26)$$

Separate factor averaging includes cross-products absent from averaging effective updates, since $(\sum_j w_j B_j)(\sum_k w_k A_k) = \sum_{j,k} w_j w_k B_j A_k$. Thus the two operations are distinct even at common rank. The product-16/sample arm also changes the stationary weights relative to Dec-LoRA, so that contrast alone does not isolate factor versus product aggregation.

To avoid materializing every dense effective update, concatenate weighted factors for a receiver as

$$L = [\sqrt{w_j \alpha / r_j} B_j]_j, \quad R = [\sqrt{w_j \alpha / r_j} A_j^\top]_j, \quad Y = LR^\top. \quad (27)$$

Thin QR gives $L = Q_L R_L$ and $R = Q_R R_R$. If $R_L R_R^\top = U \Sigma V^\top$, the best rank- r update is

$$\mathcal{C}_r(Y) = (Q_L U_{:r}) \Sigma_{:r} (Q_R V_{:r})^\top. \quad (28)$$

The stored factors divide both square-root singular-value factors by $\sqrt{\alpha/r}$, preserving Equation (24). This is algebraically the dense truncated SVD, not an additional approximate compressor. Its core dimensions grow with the sum of contributing ranks; compactness is not a universal workspace guarantee. Independent dense-oracle tests cover heterogeneous ranks, scaling, weights, gauge changes, and truncation.

The adaptive arms reuse the capability-constrained controller described above, including two warm-up rounds, a half-capability floor, smoothing, hysteresis, and quality-drop restoration. One fixed local training batch is probed at the capability ceiling. The transformer probe computes

$$s_{\text{transformer}}(G) = \frac{\|G\|_F^2}{\max\{\|G\|_2^2, 10^{-30}\}}, \quad s_{\text{transformer}}(0) = 0, \quad (29)$$

and takes the median over all adapter A and B gradients, including zero gradients. This convention differs from the earlier nonzero-gradient description: at zero- B initialization, the A gradients are zero and remain part of the transformer’s demand statistic. The score $1/(1 + \ell_i)$ supplies quality feedback. Quality affects rank decisions only in this benchmark, never the sample weights. No validation labels enter the controller. Factor-gradient stable rank is coordinate-dependent and is not assumed to estimate a uniquely required global model rank. A ceiling probe can require the ceiling adapter even when the persistent training rank is smaller.

Weighting also changes graph mixing. With $D_\pi = \text{diag}(\pi)$, the reversible similarity $S_\pi = D_\pi^{1/2} P D_\pi^{-1/2}$ is symmetric. We record the absolute spectral gap $1 - \max_{k < N} |\lambda_k(S_\pi)|$, excluding its eigenvalue one, and post-merge adapter disagreement

$$V = \sum_i \pi_i \sum_\ell \left\| X_{i\ell} - \sum_j \pi_j X_{j\ell} \right\|_F^2. \quad (30)$$

Consequently, sample versus uniform weighting is a joint objective-and-mixing intervention, not a pure reweighting with unchanged consensus speed.

Final assembly gathers serialized factor and classifier records along a breadth-first tree of the same ring. The first capacity-16 peer is selected as root before training, independently of validation scores. It forms $\hat{X}_\ell = \mathcal{C}_{16}(\sum_i \pi_i X_{i\ell})$ and $\hat{\vartheta} = \sum_i \pi_i \vartheta_i$ for product-space arms. Dec-LoRA instead assembles $\hat{A}_\ell = N^{-1} \sum_i A_{i\ell}$ and $\hat{B}_\ell = N^{-1} \sum_i B_{i\ell}$; its rank-4 comparator deploys at rank 4. Relay peers route records without intermediate adapter truncation. The assembled model is evaluated at the root and is not broadcast to smaller peers or fed back into training.

Each training message contains 24 pairs of adapter factors, the full classifier, and serialized metadata. Let H be the classifier parameter count and M_{ji}^t the measured metadata length for directed edge $j \rightarrow i$. For fp32 tensors,

$$B_{\text{train}} = \sum_{t=1}^{20} \sum_{j \rightarrow i} \left[4\{24 \cdot 1536 r_j^t + H\} + M_{ji}^t \right], \quad (31)$$

$$B_{\text{delivery}} = 320 + B_{\text{train}} + B_{\text{final gather}}. \quad (32)$$

Model messages are actually serialized, hashed, decoded, and routed inside the simulator. The 320-byte setup allowance models twenty fixed-width neighbor count records; it is not measured through that serializer. Evaluation gathers from earlier rounds are reported separately and excluded from B_{delivery} . Base-model, initial-state and graph provisioning, network framing, and subsequent dissemination are excluded. Logical per-peer gather storage, structural QR/SVD workspace, and measured whole-process CUDA peaks are different quantities. The last includes the shared backbone and multiple simulated peers and cannot establish independent-client device-memory savings.

4.7 Diagnosing label allocation, factor coordinates, and retained information

The failure analysis separates several operations that the original training loop combines. First, an IID redistribution changes only which client owns each training example, preserving every

n_i , the complete training union, the test set, and all optimizer settings. Second, a pooled-reset control inserts $\mathcal{C}_{16}$ after each epoch of an already rank-16 adapter. Its immediate effective update is unchanged up to roundoff, but its factor coordinates change. This intervention includes both Adam’s coordinate-dependent transformation and the factor-wise weight-decay geometry; it does not isolate lost moments, since moments are reset in both arms.

For the classifier, a common shift of all logits leaves the softmax unchanged. Define the class-centering matrix

$$H_C = I_C - \frac{1}{C} \mathbf{1}\mathbf{1}^\top, \quad \text{softmax}(Y\phi) = \text{softmax}(H_C Y\phi). \quad (33)$$

Centering a LoRA update is implemented by $B \leftarrow H_C B$, which cannot increase its rank. Moreover,

$$\|Y\|_F^2 = \|H_C Y\|_F^2 + \|(I_C - H_C)Y\|_F^2. \quad (34)$$

Consequently, a small discarded-energy fraction relative to $\|Y\|_F^2$ need not imply a small fraction of task-relevant centered energy. Paired controls center before SVD, and log the centered spectrum and immediate change of centered logits on a training probe. Centering after a completed projection cannot recover information already discarded.

The no-training retention diagnostic starts from $\Delta W^* = \sum_{k=1}^{16} \sigma_k u_k v_k^\top$, a completed pooled adapter. Each peer initially receives $\mathcal{C}_{r_i}(\Delta W^*)$, then repeatedly applies Equation (18) with fixed ranks and sample weights. In this aligned-basis setting, let $d_{ik} = \mathbf{1}[r_i \geq k]$ and let $c_k(t)$ contain the coefficient of $u_k v_k^\top$ at every peer. The exact-arithmetic recurrence is

$$c_k(0) = \sigma_k d_k, \quad c_k(t+1) = \text{diag}(d_k) P c_k(t), \quad \hat{c}_k(t) = \pi^\top c_k(t). \quad (35)$$

For centralized sample-weighted averaging, $P = \mathbf{1}\pi^\top$, so with $p_k = \sum_i \pi_i d_{ik}$,

$$\hat{c}_k(t) = \sigma_k p_k^{t+1}. \quad (36)$$

A component unavailable at some clients is repeatedly attenuated rather than merely omitted from those clients once. For a fixed irreducible peer kernel, modes supported by every peer persist, while modes deleted at some reachable peers decay in this no-training setting. This statement concerns fixed ranks, nonnegative aligned singular coefficients, and no local learning; it is not a theorem about the nonlinear training trajectory. Synthetic fp64 checks and measured original-basis coefficients test the recurrence against the implemented SVD and mixing kernels.

4.8 Exact synchronized-gradient positive control

To test whether separate example ownership alone precludes pooled accuracy, all peers can instead retain identical rank-16 factors $\theta_t = (A_t, B_t)$ and identical Adam state. A globally scheduled minibatch $\mathcal{B}_t$ is partitioned by the original ownership into $\mathcal{B}_{i,t}$ of size $m_{i,t}$. A neighbor-tree reduction and broadcast compute

$$g_t = \sum_{i: m_{i,t} > 0} \frac{m_{i,t}}{|\mathcal{B}_t|} \nabla_\theta \left[\frac{1}{m_{i,t}} \sum_{z \in \mathcal{B}_{i,t}} \ell(\theta_t; z) \right] = \frac{1}{|\mathcal{B}_t|} \sum_{z \in \mathcal{B}_t} \nabla_\theta \ell(\theta_t; z). \quad (37)$$

Each peer then applies the same Adam operation, including weight decay, to this common gradient. Identical initial factors and optimizer states therefore imply pooled iterates in exact arithmetic. We measure fp32 divergence rather than assuming bitwise equality, and independently check the same-state gradient identity in fp64.

The ordering of aggregation and optimization matters even before multiple local steps:

$$\sum_i \pi_i \text{AdamStep}(g_i) \neq \text{AdamStep}\left(\sum_i \pi_i g_i\right). \quad (38)$$

For a freshly initialized Adam state, its bias-corrected coordinate update is proportional to $g/(|g| + \epsilon)$, approximately $\text{sign}(g)$ away from zero; averaging signs is generally different from taking the sign of an average. Thus this positive control changes both synchronization frequency and optimizer aggregation order. It does not isolate their individual effects.

A separate one-step diagnostic directly compares these operations. For eight training minibatches per seed, both paths restart from the same rank-16 factors and zero moments, with no SVD. Let δ_{global} be the effective-update change after one Adam step on the pooled gradient, and let δ_{local} be the sample-count-weighted mean of independently optimized local effective-update changes. We compare their Frobenius cosine and relative difference,

$$\chi = \frac{\langle \delta_{\text{global}}, \delta_{\text{local}} \rangle_F}{\|\delta_{\text{global}}\|_F \|\delta_{\text{local}}\|_F}, \quad \rho = \frac{\|\delta_{\text{local}} - \delta_{\text{global}}\|_F}{\|\delta_{\text{global}}\|_F}, \quad (39)$$

and evaluate the two resulting models on that same training minibatch. This is a local arithmetic test of optimizer ordering, not a convergence or generalization experiment.

Every minibatch requires $2(N - 1)$ tree messages, each carrying $16(512 + 100) = 9792$ gradient scalars and one eight-byte count. For 11,730 fp32 steps the modeled payload is

$$B_{\text{gradient}} = 11730 \cdot 2(15 - 1) [4 \cdot 9792 + 8] \simeq 12.87 \text{ GB}. \quad (40)$$

All clients must accommodate rank 16. The simulation collapses identical peer model and Adam replicas after explicit message reduction and broadcast; its joint runtime includes both paired training paths and gradient oracles. Global minibatch scheduling, initial model dissemination, and transport framing are excluded. It is an optimization positive control, not a measured distributed-system speedup or an adaptive-rank solution.

4.9 Retained-base residual protocol

The no-training attrition mechanism motivates a distinct protocol that preserves a common global update G^t while restricting only each client’s trainable residual. At round t , client i freezes $W_0 + G^t$ and trains

$$z_i^t(x) = \left(W_0 + G^t + E_i^t\right) \phi(x) + b_0, \quad E_i^t = \frac{\alpha}{r_i^t} B_i^t A_i^t. \quad (41)$$

Each round begins with $B_i^t = 0$ and fresh seeded, nested A directions. Fixed ceilings or the controller in Equation (8) select local ranks. The existing ceiling-probe implementation is retained; because it expands and rescales the current factors, its gradient statistic depends on their coordinates as well as the frozen head. Post-training quality and domain signals use only local training probes and residual changes.

A root chosen by the deterministic rule $i_t = t \bmod N$ receives the exact weighted residual through neighbor-tree reduction. With gain η , the update and broadcast are

$$\bar{E}^t = \sum_i \pi_i^t E_i^t, \quad G^{t+1} = \mathcal{C}_{16}\left(H_C[G^t + \eta \bar{E}^t]\right), \quad W_i^{t+1} = W_0 + G^{t+1}. \quad (42)$$

The weights are sample-only, quality-only, or Equation (14). Dense numerators pass through relays without low-rank truncation. The implementation obtains the full 100×512 residual via a rank-100

SVD at the root before the single rank-16 global projection; this full-rank intermediate adds roundoff but no intended truncation. Broadcasting $W_0 + G^{t+1}$ gives every peer the same next frozen head.

If $\bar{E}^t = 0$, $H_C G^t = G^t$, and $\text{rank}(G^t) \leq 16$, Equation (42) gives $G^{t+1} = G^t$ in exact arithmetic regardless of local rank ceilings. This retention invariant addresses the repeated deletion of the accumulated model. It does not guarantee useful residual optimization. Gains $\eta \in \{1, 5, 15\}$ are a disclosed coarse search motivated by five label domains and fifteen local trajectories, not theoretically optimal rates.

This variant changes the resource model. The transient update $G^t + E_i^t$ can have rank up to $16 + r_i^t$, although deployment remains rank 16. Every peer holds the full frozen head and must perform dense aggregation and SVD when acting as root; a training-rank ceiling is not an aggregation-memory ceiling. The retained 100×512 fp32 head occupies 204,800 bytes, which can be folded into the existing frozen linear weight; an unfused implementation requires a separate buffer. Dense collective payload per round is $(N - 1)(4 \cdot 51200 + 8)$ bytes for reduction plus $(N - 1)4 \cdot 51200$ for broadcast, in addition to the weighting-control traffic. Transport buffers, SVD workspace, gradient probes, and factor-wise optimizer work remain additional costs.

4.10 Shared-model training with adaptive gradient-coordinate budgets

The synchronized positive control motivates a second variant that retains common rank- $R = 16$ factors and Adam state while computing only a subset of local factor gradients. At a minibatch, client i samples r_i rank coordinates uniformly without replacement. Let M_i keep the corresponding rows of $\nabla_A f_i$ and columns of $\nabla_B f_i$, setting the other gradient entries to zero. The importance-corrected observation is

$$\tilde{g}_i = \frac{R}{r_i} M_i g_i, \quad \mathbb{E}_{M_i}[\tilde{g}_i \mid \theta, \mathcal{B}_i, r_i] = g_i. \quad (43)$$

Unselected coordinates are missing gradient observations; they are not removed from the model. The selected gradients are computed analytically from the full rank-16 forward pass. With feature matrix X , one-hot labels Y , and mean cross-entropy logit derivative $D = (\text{softmax}(Z) - Y)/|\mathcal{B}_i|$, a selected set J gives

$$G_{A,J} = \frac{\alpha}{R} (DB_{:,J})^\top X, \quad G_{B,J} = \frac{\alpha}{R} D^\top X A_J^\top. \quad (44)$$

Autograd comparisons check these entries directly; enumeration of all coordinate subsets in a small example checks Equation (43).

At each epoch, training-probe quality and domain multipliers define

$$\mu_i = \frac{q_i \lambda_i}{\sum_j (n_j/n) q_j \lambda_j}, \quad \tilde{g}_t = \sum_i \frac{m_{i,t}}{|\mathcal{B}_t|} \mu_i \frac{R}{r_i} M_{i,t} g_{i,t}. \quad (45)$$

Sample-only training sets $\mu_i = 1$. A neighbor-tree allreduce delivers $\tilde{g}_t$ before every common Adam step. Conditional on the minibatch and current weights, its expectation over masks is the corresponding full weighted minibatch gradient. This does not imply identical Adam iterates: the mask estimator adds variance and Adam transforms its input nonlinearly.

Fixed budgets repeat (4, 8, 16); adaptive budgets use the canonical controller with an explicitly changed measurement procedure. Each peer probes a deterministic ceiling-sized subset of the shared coordinates, computes stable rank from those partial gradients, and observes the updated common model's training-probe quality after each epoch. For domain allocation, the local signal replaces a complete local-epoch update by the negative first-order probe tangent

$$u_i = -\frac{\alpha}{R} \frac{R}{R_i^{\max}} \text{vec}(B_{:,J_i} G_{A,J_i} + G_{B,J_i} A_{J_i}). \quad (46)$$

The canonical conservative allocator receives this signal and the training histogram; weights refresh once per epoch. This preserves the allocator’s function but changes its inputs and timing from the original protocol.

All peers retain the entire rank-16 model, a full forward pass, and full Adam moments. The current implementation transmits padded rank-16 gradient vectors, so it claims no communication saving. The budget r_i restricts computed gradient coordinates only; it is not a bound on model storage, optimizer storage, or total runtime. We therefore call this adaptive gradient-coordinate training rather than successful heterogeneous model-rank training. Ceiling probes, common optimizer work, and domain-control traffic are separately accounted for.

5 Experimental Design

5.1 Published-method reconstruction and quantity-skew comparison

The new study uses RoBERTa-base and SST-2 from GLUE [5, 8]. The pinned asset revisions are

Model: e2da8e2f811d1448a5b465c236feacd80ffb7b
Dataset: bcdcba79d07bc864c1c254ccfcedcce55bcc9a8c.

All 67,349 canonical training examples are used; evaluation uses the official 872 labeled validation examples, with no official-test access. The published Dec-LoRA table instead lists 66,675 training and 674 development examples. Because the source of that discrepancy is unresolved, our scores are not claimed as exact numerical reproduction or a direct win over its printed 93.81% ring result.

The reconstruction retains ten clients, a ring, rank 16, learning rate 10^{-3} , batch size 32, and 20 rounds. The equal-size anchor interprets the publication’s ambiguous $K = 1$ as one local epoch per round. Other disclosed choices are query/value adapters in all 12 blocks, $\alpha = 16$, common Gaussian A initialization with standard deviation 0.02, zero B , the complete trainable classifier, maximum token length 128 with dynamic padding, pretrained-model dropout and zero adapter dropout, and fp32 deterministic Torch operations. AdamW uses zero decay and gradient clipping at norm one, with moments reset for every peer in every round in every arm. Resetting avoids retaining stale moments across the changed SVD factor coordinates, but is a reconstruction choice rather than an attributed author setting. Rank-16 adapters contain 589,824 parameters; classifier parameters are counted separately. We do not infer the complete adapter specification from the publication’s rounded 0.60M count.

For the quantity extension, the size-ratio vector is (1, 1, 1, 1, 2, 2, 2, 4, 4, 4). Largest-remainder allocation yields exact client totals; bipartite integer rounding assigns each client/class count either the floor or ceiling of its proportional quota, preserving both client sizes and global label totals. Full source-row memberships establish disjointness and coverage. At seed s , partition randomization uses $s + 10000$, assignment of quantities to ring positions uses $s + 20000$, and an independent permutation of ceilings (4, 4, 4, 4, 8, 8, 8, 16, 16, 16) uses $s + 30000$. Matched methods reuse these assignments. Thus data quantity, rather than different topics or deliberately shifted class distributions, is the designed statistical heterogeneity.

Every quantity arm performs 211 full local updates of 32 examples per peer per round. Local streams independently reshuffle and continue across epoch boundaries, so smaller shards are revisited more often. This gives 42,200 aggregate optimizer steps and 1,350,400 example exposures per run; the equal-size full-epoch anchor instead uses 1,346,980 exposures. Equal local steps prevent larger shards from also contributing longer local trajectories before weighting. Training-probe work is additional and separately recorded. All methods use the same per-peer sample order at a paired seed, but rank-dependent computation is not identical total work.

Table 1: Seven matched quantity-skew methods. Sample weights are n_i/n ; uniform weights are $1/10$. The last four arms form the rank-policy by weighting factorial. Uniform rank 16 exceeds weaker clients’ declared adapter caps.

Arm	Local rank	Aggregation	Weights
Dec-LoRA16	16	factor average	uniform
Dec-LoRA4	4	factor average	uniform
Product16/sample	16	product + SVD	sample
Fixed/uniform	capability ceilings	product + SVD	uniform
Fixed/sample	capability ceilings	product + SVD	sample
Adaptive/uniform	controller within caps	product + SVD	uniform
Adaptive/sample	controller within caps	product + SVD	sample

After remote numerical and transport tests and seven bounded smoke runs, a fixed seed-42 screen covers all seven quantity arms and the equal-size Dec-LoRA16 anchor. All seven quantity arms are retained at seeds 43–46, irrespective of whether early scores favor the proposal. The completed campaign therefore has 35 matched quantity runs and one separate anchor. Neither a post-hoc accuracy gate nor a selected subset of seeds determines replication. The protocol is archived before full-budget results; source snapshots and hashes identify the immutable training implementation. No hyperparameter search is performed for one comparator alone. This fixes a transparent operating point but does not establish each method’s optimized performance.

Equation (5) is the primary comparison, with Dec-LoRA4 as the feasible comparator and final-round accuracy as a required secondary outcome. The factorial effects are adaptive minus fixed within each weighting choice, and sample minus uniform within each rank policy. Their difference is the interaction. These exploratory secondary intervals are not adjusted for multiple comparisons. The one-seed equal-size anchor is excluded from the five-seed aggregates and cannot by itself estimate the causal effect of quantity skew. Independent fresh-process audits recompute saved best/final predictions using separately loaded canonical data and metric code; they share the model architecture and state-install helper, so they are not an independent reimplementations of the entire transformer.

5.2 Matched pooled and integrated comparison

The primary benchmark uses all 50,000 CIFAR-100 training images and 10,000 test images. Frozen ImageNet-pretrained ResNet-18 features have dimension 512 after deterministic resizing to 224×224 and ImageNet normalization. Only LoRA factors on the random frozen 100-class head are trained. This is head-only LoRA on cached features, not end-to-end backbone or language-model fine-tuning.

Five groups of CIFAR-100 superclasses define disjoint 20-class domains. This is severe label skew, not merely a scanner or covariate shift: each client sees labels from only one of the five groups. Each domain has three clients, split by Dirichlet(0.5). The training sets and test shards are disjoint, and their unions are checked against the entire dataset. Heterogeneous arms assign repeating ceilings (4, 8, 16); uniform controls use rank 16 everywhere. A seeded permutation independent of domain ordering defines the ring, and the same graph is used by paired arms. Known domains determine the benchmark partition but do not select a privileged training hierarchy in this experiment.

All nine arms use $\alpha = 32$, learning rate 10^{-3} , weight decay 10^{-4} , minibatch size 128, one local epoch per round, 30 rounds, and seeds 42, 43, and 44. Table 2 specifies the comparators. Ordinary pooled training preserves Adam moments across epochs. Pooled-reset is an additional control that resets Adam each epoch; federated clients reset it each communication round. This separates the

Table 2: Corrected comparison. All final deployment models have rank 16. Quality and domain arms form a 2×2 ablation of rank policy and domain factors.

Arm	Training rank	Contribution weights	Training coordination
Pooled	16	individual samples	direct pooled Adam
Pooled-reset	16	individual samples	pooled Adam, reset
FedAvg-16	16 per client	n_i	central FedAvg
MH-16	16 per client	n_i	weighted peer MH
MH-sample	fixed 4/8/16	n_i	weighted peer MH
Fixed-quality	fixed 4/8/16	$n_i q_i$	weighted peer MH
Fixed-domain	fixed 4/8/16	$n_i q_i \lambda_i$	weighted peer MH
Adaptive-quality	P1 policy	$n_i q_i$	weighted peer MH
Adaptive-domain	P1 policy	$n_i q_i \lambda_i$	weighted peer MH

conventional baseline from the optimizer-reset convention used by the federation.

Each arm processes 1.5 million training-example exposures, excluding separately logged probes. Equal exposure does not mean equal optimizer steps: pooled training performs $\lceil 50000/128 \rceil$ steps per epoch, whereas the federation performs $\sum_i \lceil n_i/128 \rceil$. Counts are recorded rather than assumed equal. The pooled rank-16 model is the conventional reference, while the fixed heterogeneous arms isolate adaptation under the same client capability ceilings. They are different controls and answer different questions.

Hyperparameters, method list, seeds, and the final round are fixed before the full comparison. Test evaluation produces curves but is not used for rank decisions, weighting, early stopping, or model selection. Reported uncertainties are sample standard deviations across three seeds, not confidence intervals. Rank-policy effects are paired as adaptive minus fixed within each weighting condition; domain effects are paired as domain minus quality within each rank policy. Source hashes, feature-cache identity, initialization hashes, split manifests, phase timings, ranks, weights, matrices, transport ledgers, and final adapters are retained.

5.3 Exploratory failure analysis and residual screening

The subsequent mechanism experiments were specified after observing the corrected negative result. They therefore reuse the official test set as exploratory diagnostics, not as a fresh confirmatory test. Seven optimization controls each run 30 rounds at seeds 42–44: pooled-reset, pooled-reset with exact SVD, pooled-reset with centering and SVD, FedAvg-16, centered FedAvg-16, IID FedAvg-16, and centered IID FedAvg-16. All retain the original 50,000 training examples, sample counts, initialization, scaling, and optimizer settings. The IID arms retain the original test shards, so their personalized scores would describe legacy test strata rather than the redistributed local training populations; their main endpoint is full-test accuracy.

The no-training retention experiment has six arms and three seeds: centralized or ring mixing at uniform rank 16, repeating ranks (4, 8, 16), or repeating ranks (2, 4, 8). Every arm starts from its seed’s saved pooled adapter and is evaluated after initial distribution and 30 further project-and-mix cycles. The lower-rank schedule is a fixed stress test, not the learned adaptive controller. Rank-16 assembly measures retained information; for the all-ranks-at-most-8 stress arm it is an evaluation instrument, not a claim that one of those peers supports rank-16 deployment. The synchronized-gradient control separately compares 30 epochs of pooled and peer training at all three seeds, using the original non-IID ownership and 64 same-state fp64 gradient checks per seed. None of these diagnostics uses test-driven early stopping or chooses an intermediate checkpoint.

The retained-base candidate search instead opens only the cached original training tensor. A fixed class-stratified split reserves 50 examples per class, yielding 45,000 fitting examples and 5,000 validation examples, with split seed 20260915. Every candidate, including a newly trained conventional pooled comparator, uses this same split. Eleven seed-42 arms are specified before screening: pooled, original fixed-domain and adaptive-domain, residual uniform/sample/gain 1, residual fixed/sample/gain 1, residual adaptive/domain at gains 1, 5, and 15, residual adaptive/sample/gain 5, residual adaptive/quality/gain 5, and residual fixed/domain/gain 5. All use 30 epochs and the original Adam settings; endpoint validation accuracy, rather than an intermediate maximum, is the screening criterion.

At gain 5, adaptive versus fixed domain-weighted residuals isolate the chosen rank policy, while adaptive domain versus quality-only residuals isolate the domain multiplier. Sample versus quality-only arms separately measure quality weighting. Comparisons with the original state-averaging protocol evaluate the entire changed recipe, which also alters synchronization, retained capacity, centering, and factor initialization. The finite gains are an exploratory hyperparameter search.

After observing the retained-base screening outcomes, a second prospective plan specifies six shared-model partial-gradient arms on the same 45,000/5,000 split: uniform/sample as a driver control, fixed/sample, adaptive/sample, adaptive/quality, fixed/domain, and adaptive/domain. The initial screen again uses seed 42, 30 epochs, and the original learning rate, decay, batch size, and scaling. Before that screen finishes, a separate plan selects fixed/domain, adaptive/domain, and the uniform driver control for replication at seeds 43 and 44, because these arms address the target combined method and its capability-matched comparator rather than because one has the best completed endpoint. Conventional pooled LoRA is also trained on the same 45,000 examples at both replication seeds. Thus pooled, uniform driver, fixed/domain, and adaptive/domain have paired three-seed validation comparisons; the remaining partial-gradient contrasts remain single-seed exploration.

This sequence is an explicitly adaptive research investigation; the second variant was not hypothesized before the first screen. Neither candidate stage opens the official test tensor, including the replications. Their reported outcomes are validation accuracy, and no noninferiority or equivalence margin is specified. The use of a separate fitting/validation split does not erase the earlier official-test observations that motivated the investigation. Every attempted run, failure, source snapshot, split, per-round observation, and final checkpoint is retained.

5.4 Implementation validation and excluded pilots

Validation checks scaled heterogeneous merging, weighted stationarity and detailed balance, graph-restricted traffic, trainable rank expansion, deterministic training probes, and final peer assembly against a direct weighted-SVD test oracle. Paired smoke runs check that adaptive probes leave fixed/adaptive warm-up trajectories unchanged and that pooled and pooled-reset have identical first-epoch behavior. The direct weighted average is used to validate the reduction, not to replace it at evaluation.

An audit of earlier pilot results found issues that affect their interpretation. First, the old weighting wrapper computed Sinkhorn(DWD) for an already doubly-stochastic W and positive diagonal D . Since diagonal rescaling is precisely what Sinkhorn balancing removes,

$$\text{Sinkhorn}(DWD) = W \tag{47}$$

under the usual uniqueness conditions, up to numerical tolerance. Consequently, identical weighted and unweighted pilot outcomes do not establish that domain weighting is ineffective or preserves accuracy: the intended intervention was canceled.

Second, the pilot’s so-called adaptive-rank arm used a median-loss threshold and half-ceiling rule, not the Project 1 gradient-based controller. Its measured performance describes that heuristic only. Third, the separate pooled pilot hardcoded $\alpha = 32$ while the compared decentralized run used $\alpha = 16$; it also contrasted a sample-based pooled objective with client-uniform gossip and an evaluator-assembled consensus adapter. These are not matched tests of Equation (4). We retain their raw artifacts for provenance but exclude their success/failure and weighting-neutrality conclusions. The corrected experiment uses one common scaling constant, the intended policies, explicit objective controls, and actual peer assembly.

5.5 Supporting component studies

The earlier Project 3 protocol battery used 50 rounds, the same three seeds, and uniform rank 16 or heterogeneous ranks (4, 12, 32). It compared client-uniform local, FedAvg, flat gossip, and an oracle hierarchy. This objective, rank schedule, and round budget differ from the corrected comparison. Historical consensus scores use a centrally computed uniform mean followed by a reference-rank SVD; they are diagnostic scores, not outputs of the new final tree reduction.

The oracle uses known domains and the matrix $W^t = W_{\text{intra}}W_{\text{bridge}}W_{\text{intra}}$ on bridge rounds, with intra-domain mixing otherwise. Its separate online-discovery extension projects effective updates into 64 dimensions, normalizes them, and smooths them:

$$z_i^t = \frac{P \text{vec}(\Delta W_i^t)}{\|P \text{vec}(\Delta W_i^t)\|_2}, \quad h_i^t = \frac{0.8h_i^{t-1} + 0.2z_i^t}{\|0.8h_i^{t-1} + 0.2z_i^t\|_2}. \quad (48)$$

With affinity $a_{ij} = h_i^\top h_j$, a graph-supported kernel $K_{ij} = \mathbf{1}_{(i,j) \in E \cup I} \exp[(a_{ij} - a_{\max})/\tau]$ is Sinkhorn-balanced and blended with a self-weight floor. This pairwise-affinity kernel differs from the canceled clientwise diagonal weighting above. Cluster count is chosen by maximum silhouette over $k \in \{2, \dots, \min(8, N - 1)\}$; nonpositive scores select one group. Labels are used only to score discovery. The historical callback observes all adapters centrally, so this discovery result is not a neighborhood-local protocol and is not included in the corrected fixed-ring ablation.

Project 1’s original five-task battery compared a controller against fixed rank 32, above its capability ceilings. A later Fashion-MNIST rerun used a feasible fixed (4, 8, 16) reference and the revised controller. Project 2 separately evaluated recorded leave-one-client-out contributions, $y_i = \text{Acc}(\text{all}) - \text{Acc}(\text{all} \setminus i)$, on 75 real-data observations. Its exploratory fitted score used ridge regression, $\hat{\theta}_\kappa = (X^\top X + \kappa I)^{-1} X^\top y$, with leave-one-task-out selection of κ . The corrected training experiment uses the conservative unfitted allocator in Equation (13), not a fit to held-out test accuracy.

6 Results

6.1 Quantity skew: a functioning protocol without a demonstrated win

All 36 full-budget runs complete: seven quantity arms at five matched seeds plus the equal-size anchor. The separate evaluator reproduces the exact raw prediction vectors of all 72 saved best/final checkpoints and verifies their source, data, and checkpoint hashes. Table 3 and Figure 1 report all seven methods, without mixing smoke results or the one-seed anchor into their uncertainty estimates.

Adaptive/sample reaches $94.24 \pm 0.25\%$ best validation accuracy, versus $94.54 \pm 0.24\%$ for reconstructed Dec-LoRA16. The paired primary difference is -0.298 ± 0.368 percentage points,

Table 3: Completed SST-2 quantity-skew comparison at seeds 42–46. Accuracy is percent, mean $\pm$ sample SD. Best over 20 rounds is primary; round 20 is secondary. Mean traffic is decimal GB. Training includes classifier and metadata; delivery adds modeled setup and one final gather, excluding provisioning. A dagger marks references exceeding some peers’ adapter caps.

Method	Best validation	Final validation	Train GB	Delivery GB
Dec-LoRA16 [†]	94.54 $\pm$ 0.24	94.45 $\pm$ 0.32	1.893	2.011
Dec-LoRA4	94.50 $\pm$ 0.33	94.27 $\pm$ 0.37	1.185	1.259
Product16/sample [†]	94.72 $\pm$ 0.18	94.43 $\pm$ 0.22	1.893	2.011
Fixed/uniform	94.52 $\pm$ 0.26	94.20 $\pm$ 0.32	1.468	1.556
Fixed/sample	94.40 $\pm$ 0.13	94.22 $\pm$ 0.24	1.468	1.556
Adaptive/uniform	94.17 $\pm$ 0.22	93.81 $\pm$ 0.51	1.296	1.373
Adaptive/sample	94.24 $\pm$ 0.25	93.90 $\pm$ 0.19	1.291	1.368

with unadjusted 95% t interval $[-0.755, 0.159]$. The interval includes zero, but this is not an equivalence or noninferiority result. Its final-round difference is -0.550 ± 0.297 points, with interval $[-0.919, -0.182]$. Reporting only the selected best scores would obscure this less favorable final endpoint. The feasible Dec-LoRA4 comparator also has a higher mean best score, with adaptive minus rank-4 difference -0.252 ± 0.348 points and interval $[-0.684, 0.180]$. These results support neither superiority nor established parity for the proposed combined method.

The product-16/sample control has the highest mean best score, $94.72 \pm 0.18\%$; adaptive/sample trails it by 0.482 points, with difference interval $[-0.877, -0.087]$. Its final score, $94.43 \pm 0.22\%$, is similar in magnitude to Dec-LoRA16’s, illustrating the influence of the best-round endpoint. This arm exceeds weaker clients’ rank ceilings and changes both aggregation and weighting relative to Dec-LoRA. It cannot be used to attribute its score to one operation alone or to claim a resource-feasible solution.

6.2 Rank and sample-weight effects in the quantity benchmark

At uniform weights, adaptation changes best accuracy by -0.344 points, with a 95% interval of $[-0.569, -0.119]$. At sample weights, its effect is -0.161 points, with interval $[-0.399, 0.078]$. Sample weighting changes fixed-rank accuracy by -0.115 points, with interval $[-0.340, 0.110]$. Its effect at adaptive ranks is $+0.069$ points, with interval $[-0.169, 0.307]$.

The adaptation main effect is -0.252 points, with interval $[-0.382, -0.123]$. The weighting main effect is -0.023 points, with interval $[-0.152, 0.106]$. Their interaction is $+0.183$ points, with interval $[-0.201, 0.568]$. These paired descriptive contrasts (Figure 2) show no demonstrated weighting benefit and a less favorable accuracy pattern for this adaptive policy. Small or nonsignificant effects do not prove that weighting is universally neutral. Counts, capacities, and ring positions vary jointly across only five seeds, and nonuniform weights also change the mixing kernel.

Adaptive/sample has a mean persistent training rank of 5.792 ± 0.094 across the five runs, compared with mean ceiling rank 8.8 in fixed heterogeneous training. Its final mean client ranks are 5.6, 5.0, 5.2, 5.4, and 5.4 at seeds 42–46. Figure 3 relates these rank trajectories to validation learning curves. Every adaptive run adds 12,800 probe-example evaluations: half are backward ceiling-rank demand probes and half forward quality checks. Those costs, the trainable classifier, the full backbone forward/backward path, and the root’s assembly workspace prevent converting a rank reduction directly into a total-compute or device-memory saving.

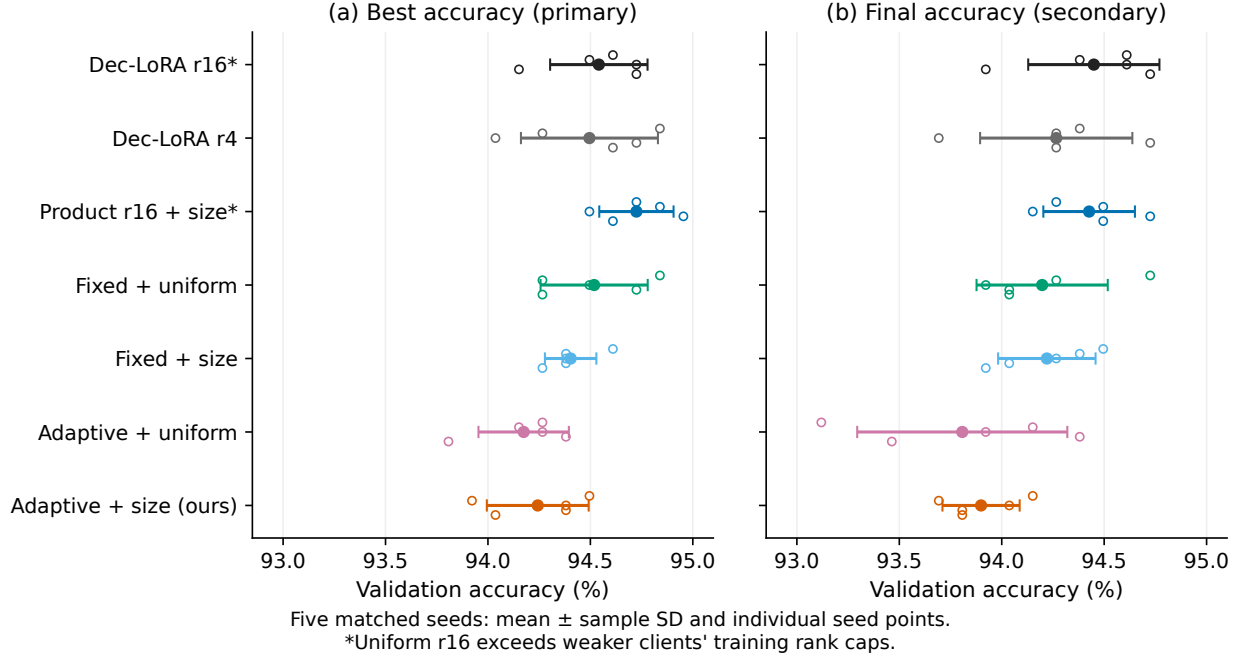


Figure 1: All seven full-budget quantity-skew arms on official SST-2 validation data. Individual seeds, means, and sample SD distinguish the primary best-round endpoint from final-round performance. Rank-16 references exceed weaker peers’ declared adapter caps; the equal-size anchor and smoke runs are excluded.

6.3 Quantity-skew communication tradeoff and reconstruction anchor

Adaptive/sample transmits a mean 1.291 GB during training, 31.81% below Dec-LoRA16’s 1.893 GB and 12.08% below fixed/sample’s 1.468 GB. However, it uses 8.92% more training traffic than feasible Dec-LoRA4’s 1.185 GB while also having a lower mean accuracy. Thus it offers a measured traffic-versus-accuracy tradeoff relative to rank 16, but does not dominate the simpler feasible rank-4 baseline. The classifier alone contributes 0.947 GB to every run’s training exchange. LoRA-factor-only savings therefore materially exaggerate total model-message savings. Figure 4 retains all methods and explicitly marks the unconstrained rank-16 references.

The equal-size Dec-LoRA16 anchor reaches 94.15% best and final validation accuracy at seed 42. It is a useful reconstruction check but has only one seed and slightly different sample exposure from the fixed-step quantity extension. The number is not evidence that we exceeded the publication’s 93.81% result: the canonical split, disclosed implementation choices, and uncertain local-update interpretation differ. All scores in the new study concern one public validation set, one model family, one topology, and one quantity-ratio schedule.

6.4 Corrected pooled comparison

The corrected experiment does not meet the pooled-accuracy criterion (Table 4). Ordinary pooled LoRA reaches $57.23 \pm 0.59\%$, compared with $6.90 \pm 2.03\%$ for the complete adaptive-domain arm: a seed-paired difference of -50.33 ± 2.46 percentage points. Every federated arm trails the pooled reference in every seed. This is a measured negative result for the specified implementation and training budget, not an inference from the previously mismatched pilots.

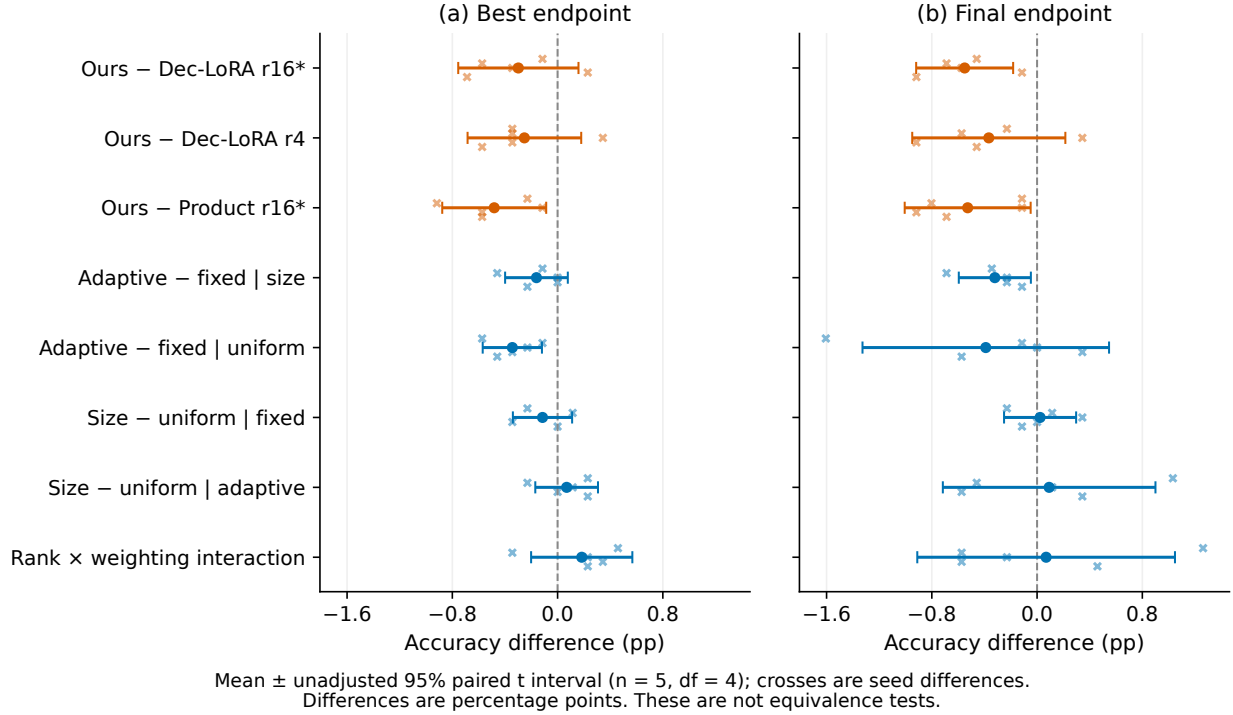


Figure 2: Seed-paired comparative and factorial effects in the quantity benchmark, in percentage points. Each method uses its own selected best round for primary contrasts. Intervals are unadjusted 95% Student- t intervals over five seeds, not equivalence tests or uncertainty on a new hidden-test population.

The shortfall is not confined to adaptive rank. Uniform-rank FedAvg reaches $20.06 \pm 0.44\%$, uniform-rank peer MH reaches $12.69 \pm 3.55\%$, and heterogeneous sample-weighted MH reaches $7.71 \pm 0.81\%$. Pooled-reset remains close to ordinary pooled training at $57.25 \pm 0.66\%$, with a paired difference of $+0.03 \pm 0.07$ points. Thus resetting Adam in pooled training does not reproduce the much larger federated shortfall, although this control does not isolate optimizer behavior under local domain shifts and repeated factor refactorization.

The factorial comparison isolates the selected policies more directly (Table 5). Adaptive-quality reaches $6.48 \pm 1.67\%$ versus $10.69 \pm 0.73\%$ for fixed-quality, a paired change of -4.21 ± 1.79 points. With domain weights, the corresponding change is -3.89 ± 1.44 points against fixed-domain’s $10.78 \pm 2.01\%$. Domain factors change accuracy by $+0.10 \pm 1.63$ points at fixed ranks and $+0.42 \pm 0.38$ points under adaptive ranks. Unlike the canceled pilot operation, this weighting intervention changes the actual mixing matrices and trajectories. Its small observed effects do not restore pooled accuracy; a claim of broad no-regression is also unwarranted from three seeds, particularly given the variability at fixed ranks. No improvement over conventional training was required for success.

Rank adaptation reduces mean training-factor payload from 20.56 MB to 11.96 MB in quality-only training, and to 11.95 MB with domain weighting, approximately 41.9% in either condition. Including the required control traffic and final assembly, quality-only deployment payload falls from 23.62 MB to 15.01 MB, a 36.4% reduction accompanied by the measured accuracy loss. Domain-control exchange alone costs 3.275 GB per run; its total deployment payload is 3.299 GB at fixed ranks and 3.290 GB with adaptation, only a 0.26% reduction. MB and GB here denote 10^6

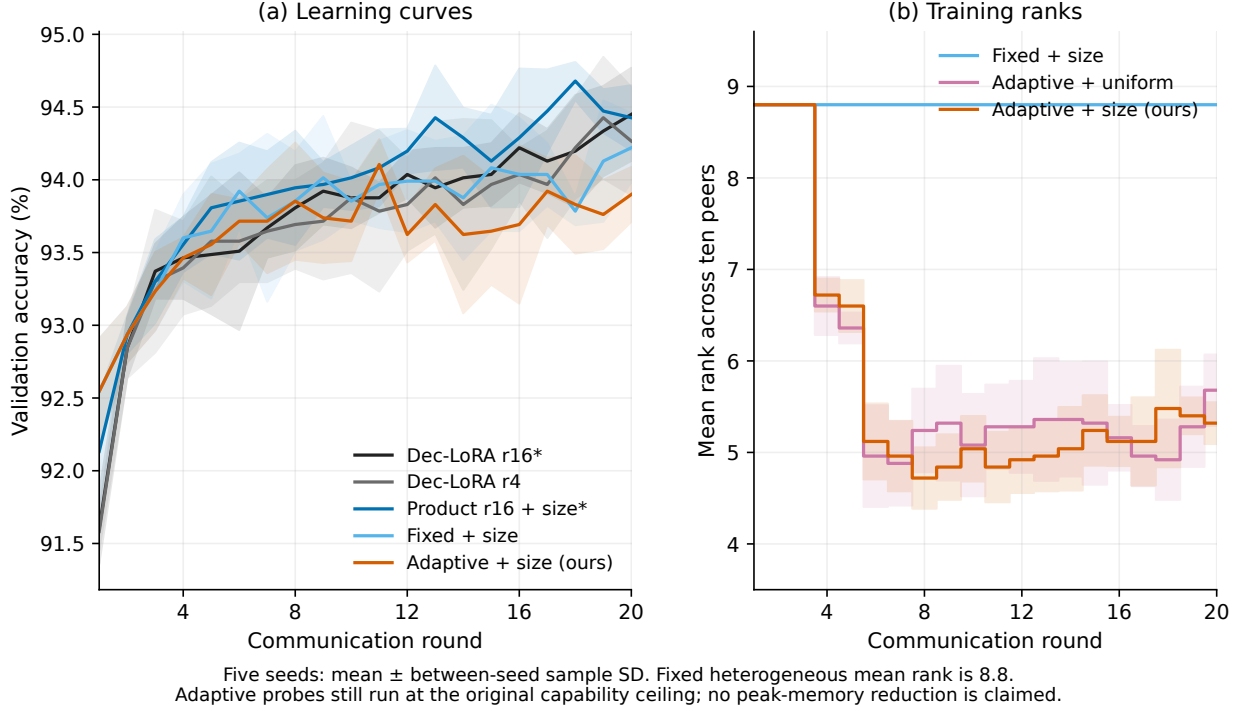


Figure 3: Validation learning curves and mean client training rank over the fixed 20-round quantity-skew budget. Curves summarize five paired seeds; evaluation assemblies do not alter subsequent peer training. Training rank excludes ceiling probes and root assembly requirements.

and 10^9 bytes. Each adaptive run additionally processes 57,600 gradient-probe examples and 57,600 quality-probe examples. These costs preclude interpreting the factor reduction as a total-compute saving.

All 27 runs process the same 1.5 million training-example exposures. They record 11,730 pooled Adam steps and 12,000 aggregate client Adam steps, with the latter determined by client minibatch boundaries. Independent evaluation of every saved checkpoint reproduces all 27 recorded fp32 correct counts on all 10,000 examples, and verifies rank-16 final updates and prediction hashes. A separate fp64 factorized forward pass disagrees on four near-tied predictions across 270,000 evaluations. Only one changes a correct count: seed-43 adaptive-quality increases by one of 10,000 examples, or 0.01 percentage points, with a top-two fp64 logit margin of 5.96×10^{-8} . The original fp32 results and the initial strict cross-precision check are retained. This precision sensitivity does not change the reported conclusion.

6.5 Label allocation explains much of the uniform-rank shortfall

The unmodified pooled-reset and FedAvg controls reproduce the earlier scientific trajectories at all three seeds. Table 6 and Figure 10 report the completed paired interventions. Exact rank-16 SVD after each pooled epoch changes mean final accuracy from 57.25% to 57.17%; adding class centering also gives 57.17%. The maximum immediate centered-logit discrepancy on the training probes is about 4.01×10^{-5} . These controls do not support factor refactorization alone as the explanation for the roughly 37-point pooled-FedAvg gap.

Redistributing the same training examples IID raises FedAvg from $20.06 \pm 0.44\%$ to $53.25 \pm 0.44\%$,

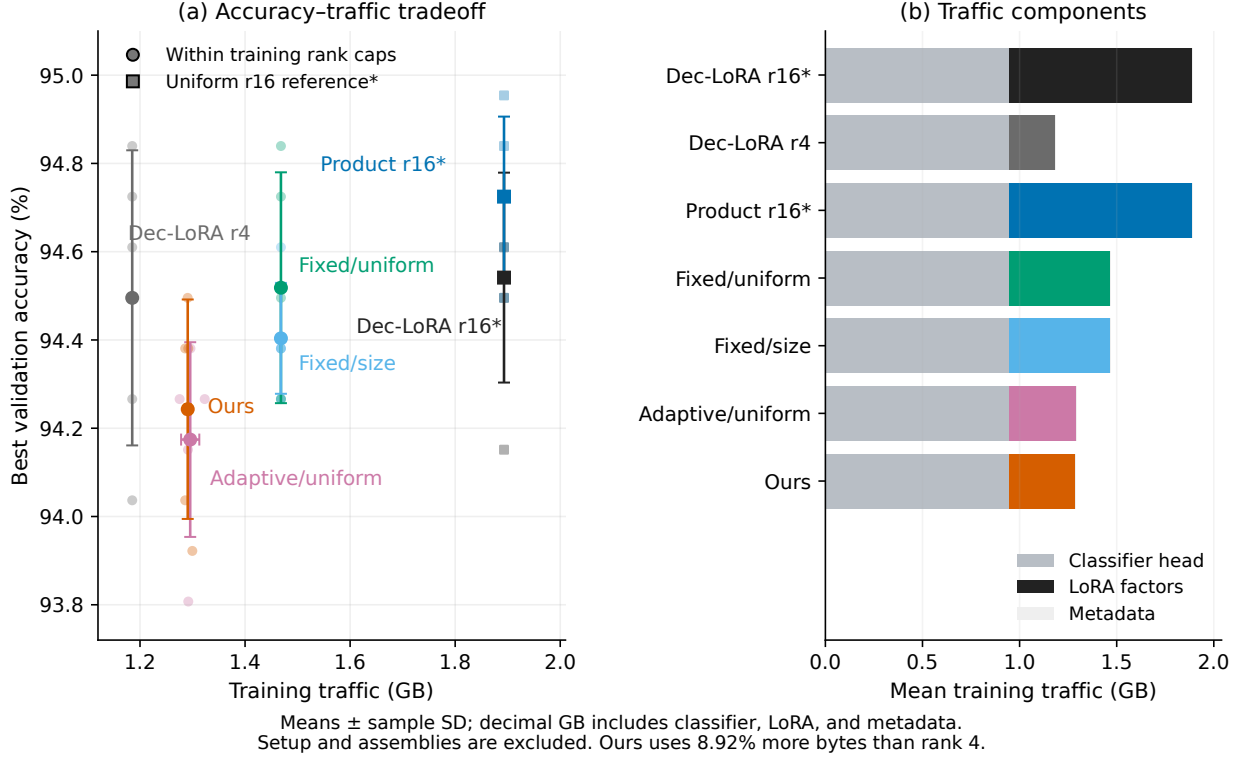


Figure 4: Best SST-2 validation accuracy against serialized training traffic. Traffic includes adapter factors, full classifier parameters, and metadata; it excludes provisioning, modeled setup, evaluation gathers, and final delivery. The rank-4 baseline is feasible at every declared adapter cap. This plot establishes neither independent-device memory savings nor a measured network speedup.

a paired increase of 33.20 ± 0.84 percentage points. A paired gap of 4.00 ± 1.10 points to pooled-reset remains. Thus severe label allocation is a major contributor under this protocol, but removing it does not demonstrate pooled parity. It also changes the original decentralized learning problem rather than providing a remedy for participants whose domains cannot be redistributed.

Class centering does not repair non-IID FedAvg: its accuracy falls to $18.51 \pm 1.24\%$, while IID centering gives $53.30 \pm 0.27\%$. The mean pre-aggregation common-logit energy fraction is 96.45% in original FedAvg versus 2.07% under IID ownership; this helps explain why an uncentered Frobenius denominator can obscure the geometry of the predictive update. However, the centering intervention itself does not recover accuracy. Common-mode energy is therefore a diagnostic, not a demonstrated standalone cause or successful correction.

6.6 Heterogeneous projection can destroy a learned model without training

The 18 no-training runs support the information-loss mechanism in Equations (35) and (36). Uniform-rank mixing preserves the pooled model’s $57.23 \pm 0.59\%$ accuracy over 30 cycles. In contrast, initially projecting the same oracle adapter to ranks (4, 8, 16) and assembling it gives $48.03 \pm 0.83\%$; repeated ring mixing and receiver projection reduce this to $13.53 \pm 2.32\%$ without a single optimizer step. The additional loss after initial distribution is 34.50 ± 2.75 points. The lower-rank (2, 4, 8) stress schedule falls from $21.11 \pm 1.08\%$ to $5.85 \pm 1.63\%$.

Centralized heterogeneous averaging exhibits the same attrition (Table 7); replacing the ring

Arm	Full-test accuracy (%)	Paired gap to pooled (pp)	Seeds
Pooled rank 16	57.23 ± 0.59	$+0.00 \pm 0.00$	3
Pooled rank 16; Adam reset	57.25 ± 0.66	$+0.03 \pm 0.07$	3
FedAvg rank 16	20.06 ± 0.44	-37.17 ± 0.40	3
Peer rank 16	12.69 ± 3.55	-44.54 ± 4.12	3
Peer ranks 4/8/16; sample weights	7.71 ± 0.81	-49.51 ± 1.38	3
Fixed ranks; quality weights	10.69 ± 0.73	-46.54 ± 0.47	3
Fixed ranks; domain weights	10.78 ± 2.01	-46.44 ± 2.08	3
Adaptive ranks; quality weights	6.48 ± 1.67	-50.75 ± 2.08	3
Adaptive ranks; domain weights	6.90 ± 2.03	-50.33 ± 2.46	3

Table 4: Corrected integrated comparison on the full CIFAR-100 test set. Each entry is the mean $\pm$ sample standard deviation across seeds. Differences are calculated within matching seeds against pooled rank-16 LoRA with persistent Adam moments. Peer results evaluate the final rank-16 assembly. The backbone is frozen; peer communication is simulated in one process. No noninferiority or equivalence margin was prespecified.

Paired contrast	Accuracy difference (pp)
Adaptive minus fixed; quality only	-4.21 ± 1.79
Adaptive minus fixed; domain weighted	-3.89 ± 1.44
Domain minus quality; fixed ranks	$+0.10 \pm 1.63$
Domain minus quality; adaptive ranks	$+0.42 \pm 0.38$
Rank $\times$ domain interaction	$+0.32 \pm 1.35$

Table 5: Paired factorial contrasts, mean $\pm$ sample standard deviation across seeds. Positive values favor the first condition. All four arms use the same graph and sample-times-quality base allocation. The interaction is $(A_D - F_D) - (A_Q - F_Q)$. These are descriptive contrasts, not equivalence tests.

with exact averaging therefore does not by itself preserve the accumulated model when every receiver repeatedly truncates it to its own capacity. After 30 ring cycles, the (4, 8, 16) and (2, 4, 8) schedules retain 42.78% and 25.78% of the oracle adapter’s energy, respectively. The comparison starts from an unavailable pooled oracle and measures robustness of an already learned state, not a federated training result. It establishes one concrete failure mechanism, without attributing the entire original accuracy gap to that mechanism.

6.7 Synchronized peer gradients match the pooled reference

Exact neighbor-tree gradient synchronization reaches the same $57.23 \pm 0.59\%$ final accuracy as conventional pooled LoRA: the paired endpoints are 57.80%, 56.63%, and 57.25% for seeds 42–44. All test predictions agree between the paired paths at every one of the 90 epoch evaluations (Figure 12). The factors are not bitwise identical: the largest final factor discrepancy is 5.36×10^{-7} . All 192 same-state fp64 gradient checks pass, with maximum absolute error 3.33×10^{-16} . Independent reconstruction of the forward pass reproduces all six saved checkpoint counts and prediction hashes.

This is successful numerical reproduction of the specified pooled learning path under separate data ownership and graph-restricted gradient messages. It costs 328,440 messages and 12.87 GB per run, requires rank 16 at every peer, and uses globally coordinated minibatches with common optimizer state. It therefore shows that decentralized data ownership itself need not produce the original accuracy loss, while leaving the heterogeneous-rank resource objective unresolved. It is not a claim of statistical equivalence across other tasks or of privacy.

The one-step ordering diagnostic further distinguishes gradient aggregation from aggregation

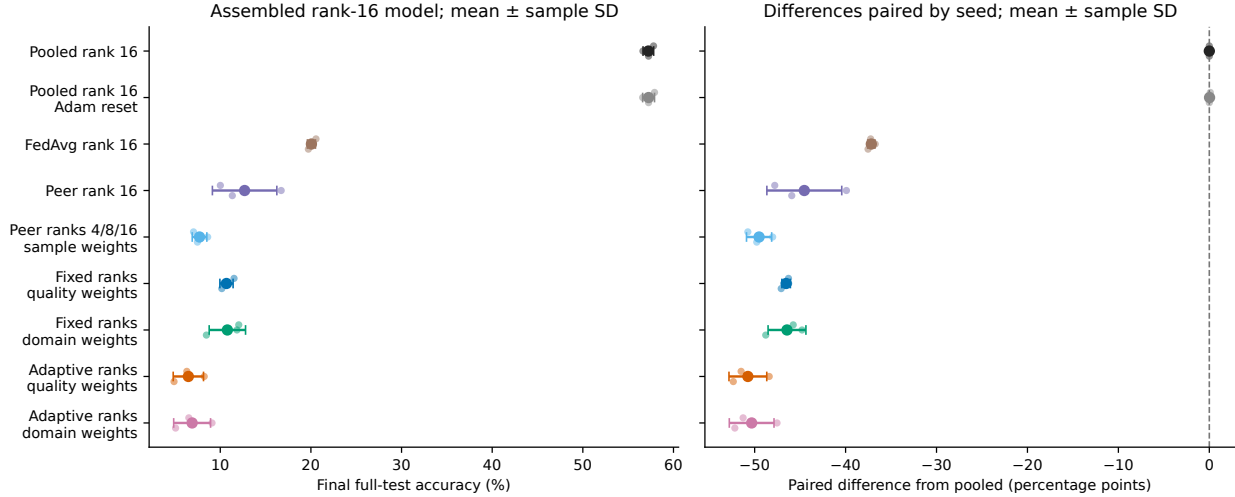


Figure 5: Corrected final rank-16 full-test comparison. All methods use the same test set; error bars show sample standard deviation across seeds 42–44. Ordinary pooled LoRA is the primary reference.

Table 6: Exploratory optimization controls after 30 rounds, using all 50,000 training and 10,000 test examples. Values are full-test accuracy in percent, mean $\pm$ sample SD across seeds 42–44. IID changes ownership while preserving every client’s sample count.

Control	Full-test accuracy
Pooled-reset	57.25 ± 0.66
Pooled-reset + exact SVD	57.17 ± 0.56
Pooled-reset + centering + exact SVD	57.17 ± 0.49
FedAvg-16, original label skew	20.06 ± 0.44
FedAvg-16 + centering	18.51 ± 1.24
FedAvg-16, IID ownership	53.25 ± 0.44
FedAvg-16, IID + centering	53.30 ± 0.27

after local Adam. Across 24 training minibatches, weighted local gradients agree with the pooled gradient to 6.94×10^{-17} , but their independently optimized effective updates have mean cosine $\chi = 0.242$ with the pooled Adam update and mean relative difference $\rho = 1.104$. Mean same-minibatch cross entropy falls by 0.01534 after the pooled-gradient Adam step and by 0.00144 after averaging the independent local-Adam updates. No SVD, multi-step drift, or test examples enter this diagnostic. It confirms that optimizer ordering is already consequential at one step; it does not quantify its share of the full 30-round accuracy gap.

6.8 Retaining a common base does not rescue the screened residual recipes

All eight retained-base residual candidates fail the seed-42 validation screen (Figure 13). Their final accuracies range from 1.42% to 2.88%, compared with 57.64% for conventional LoRA on the same 45,000 fitting examples and 5,000 validation examples. The original fixed-domain and adaptive-domain controls reach 4.52% and 4.62%, respectively, on this split. These validation values have a different training budget and evaluation set from the primary full-data experiment and should not be compared as paired improvements to its endpoints.

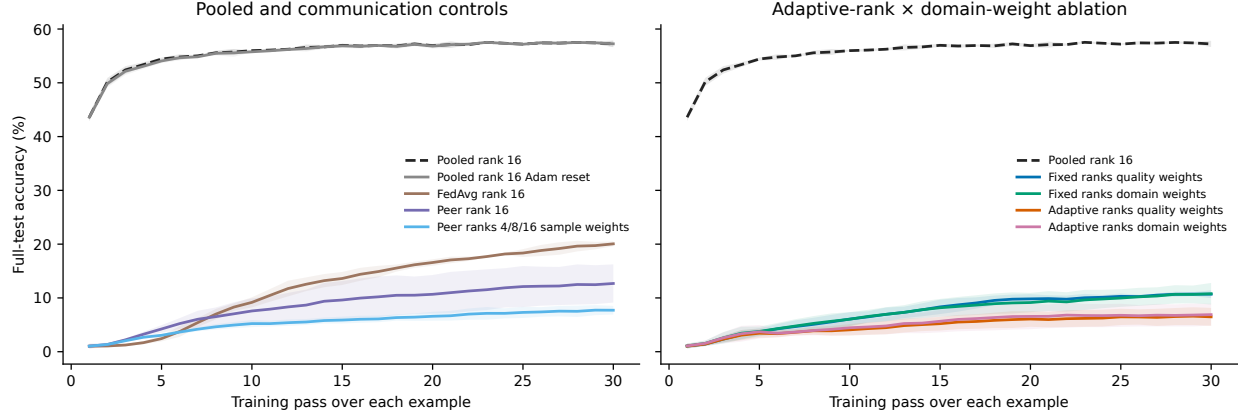


Figure 6: Full-test accuracy over the fixed 30-round budget. Each curve evaluates the actual deployment model at that round. Intermediate peer assembly supports evaluation only and does not feed a global model back into decentralized training.

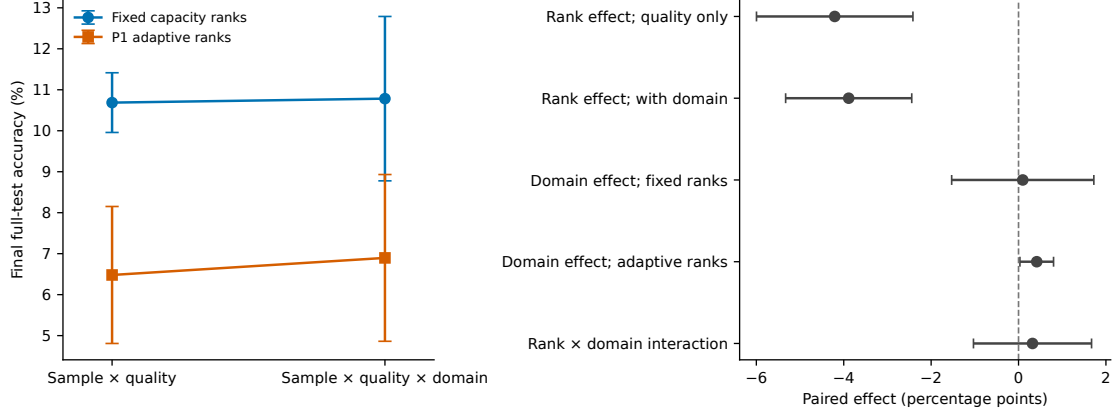


Figure 7: Paired component effects within the corrected experiment. Rank adaptation is compared at the same weighting condition; domain factors are compared at the same rank policy. These comparisons complement the primary comparison against pooled training.

The residual adaptive/domain recipe reaches 1.74%, 2.64%, and 1.48% at gains 1, 5, and 15. At gain 5, fixed/domain reaches 2.60%, adaptive/sample 2.88%, and adaptive/quality 2.46%. Uniform/sample and fixed/sample at gain 1 reach 1.68% and 1.42%. Thus none of the disclosed residual choices approaches the pooled reference. The gain-15 model remains finite but exhibits severe loss growth: local training cross entropy rises from 3.520 to 47.848, with a maximum of 53.490 and independently reconstructed validation cross entropy of 41.463. This is not a NaN failure or a proof of asymptotic divergence.

The zero-residual retention invariant holds algebraically, but preserving past state does not ensure that newly optimized residuals provide useful global learning. These results reject the eight tested recipes at one seed and budget rather than retained-state methods in general. Independent direct fp32 reconstruction reproduces all 11 saved checkpoint counts, including the unsuccessful and high-gain cases. The dense residual reduction and head broadcast require 164.07 MiB per 30-epoch run before quality/domain control; adding domain control raises the total to 3287.52 MiB. Smaller trainable residuals therefore neither establish the original complete-model capacity restriction nor

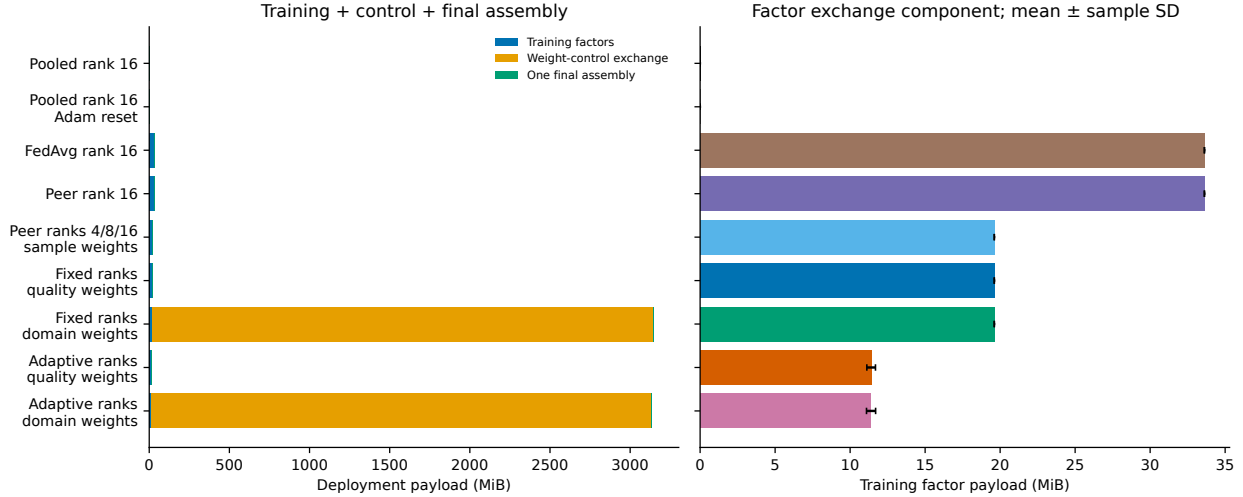


Figure 8: Simulated deployment payload: factor exchange during training, all weighting-control traffic, and one final peer assembly. Dense control information is included. These counts are not measured bandwidth or network latency.

Table 7: Oracle-initialized information-retention test with zero training. Accuracy is measured after initial per-peer rank projection and assembly, then after 30 further mixing/projection cycles. Values are percent, mean $\pm$ sample SD over three pooled checkpoints.

Mixing	Fixed peer ranks	Initial assembly	After 30 cycles
Ring	16	57.23 ± 0.59	57.23 ± 0.59
Ring	4/8/16	48.03 ± 0.83	13.53 ± 2.32
Ring	2/4/8	21.11 ± 1.08	5.85 ± 1.63
Central	16	57.23 ± 0.59	57.23 ± 0.59
Central	4/8/16	48.03 ± 0.83	11.27 ± 0.92
Central	2/4/8	21.11 ± 1.08	4.46 ± 0.24

demonstrate an efficient repair.

6.9 Partial gradients approach the reference under a different capacity model

The shared-model variant gives a substantially stronger validation result (Table 8 and Figure 14). Conventional LoRA and the uniform shared-gradient driver control both reach $57.18 \pm 0.71\%$ across seeds 42–44. Their independently reconstructed final prediction vectors are identical at every seed. Recorded intermediate correct counts match in 89 of 90 epoch evaluations: at seed 42, epoch 1, the uniform driver has 2058 correct predictions versus 2059 for pooled training. This validation-only analytic-gradient implementation is distinct from the earlier full-data synchronized-gradient control, whose predictions match at all 90 evaluations. Equal final predictions do not imply bitwise-identical optimization trajectories.

Fixed/domain partial gradients reach $55.90 \pm 0.59\%$, while adaptive/domain reaches $55.03 \pm 1.01\%$. The paired adaptive-minus-fixed difference is -0.87 ± 0.86 percentage points. The adaptive recipe is below its fixed-budget comparator at all three seeds; it does not establish that adaptation preserves accuracy. Neither the remaining pooled gap nor its variation establishes equivalence or noninferiority without a specified margin and appropriate design.

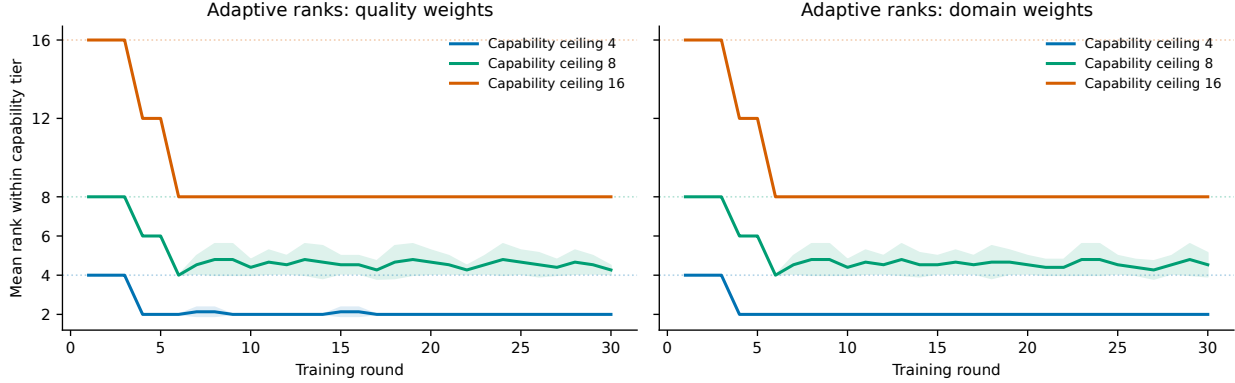


Figure 9: Adaptive ranks under the published capability ceilings. Rank choices and gradient probes are recorded for each client; rank reductions alone do not account for probe, SVD, or metadata costs.

Table 8: Final validation accuracy on the common 45,000/5,000 training split. Values are percent, mean $\pm$ sample SD across three paired seeds. Differences use conventional pooled LoRA on this same split as reference. Partial gradients constrain computed coordinates, not complete-model rank.

Method	Validation accuracy	Paired difference (pp)
Conventional pooled LoRA	57.18 ± 0.71	—
Uniform shared-gradient control	57.18 ± 0.71	0.00 ± 0.00
Fixed partial gradients + domain	55.90 ± 0.59	-1.28 ± 1.24
Adaptive partial gradients + domain	55.03 ± 1.01	-2.15 ± 1.69

The remaining screening contrasts have only seed 42: fixed/sample reaches 54.96%, adaptive/sample 54.68%, adaptive/quality 54.52%, fixed/domain 55.26%, and adaptive/domain 54.80%, against 57.64% for pooled and uniform controls. All are retained, rather than selecting only the best endpoint. The domain addition changes adaptive quality-only accuracy by 0.28 points in this single-seed screen; this does not establish a general benefit of domain allocation.

Independent evaluation reproduces all 15 partial-gradient and pooled checkpoint counts. Each run processes 1,350,000 training-example exposures and 10,560 common optimizer steps. Full rank-16 forward computation, complete factors and Adam moments, and padded rank-16 gradient transport remain required. Reduced rank-sample products measure selected gradient-coordinate work only and exclude probes, forward computation, optimizer work and aggregation. This variant shows that a changed optimization protocol can narrow the observed accuracy gap, but does not satisfy the original heterogeneous complete-model memory claim or establish privacy.

6.10 What the component evidence establishes

Table 9 retains the earlier 50-round protocol results. With uniform rank 16, the oracle hierarchy reaches $68.98 \pm 0.95\%$ personalized accuracy, while its diagnostic consensus accuracy is $10.25 \pm 0.62\%$. This contrast illustrates why a domain-specific personalized score cannot establish parity of one globally deployed model. The oracle result remains useful evidence that known domain structure changes the behavior of local models.

Under the historical heterogeneous schedule, factor zero-padding gives personalized accuracy of $4.71 \pm 0.22\%$ for FedAvg, $20.63 \pm 4.40\%$ for MH, and $31.24 \pm 1.27\%$ for the oracle hierarchy, compared

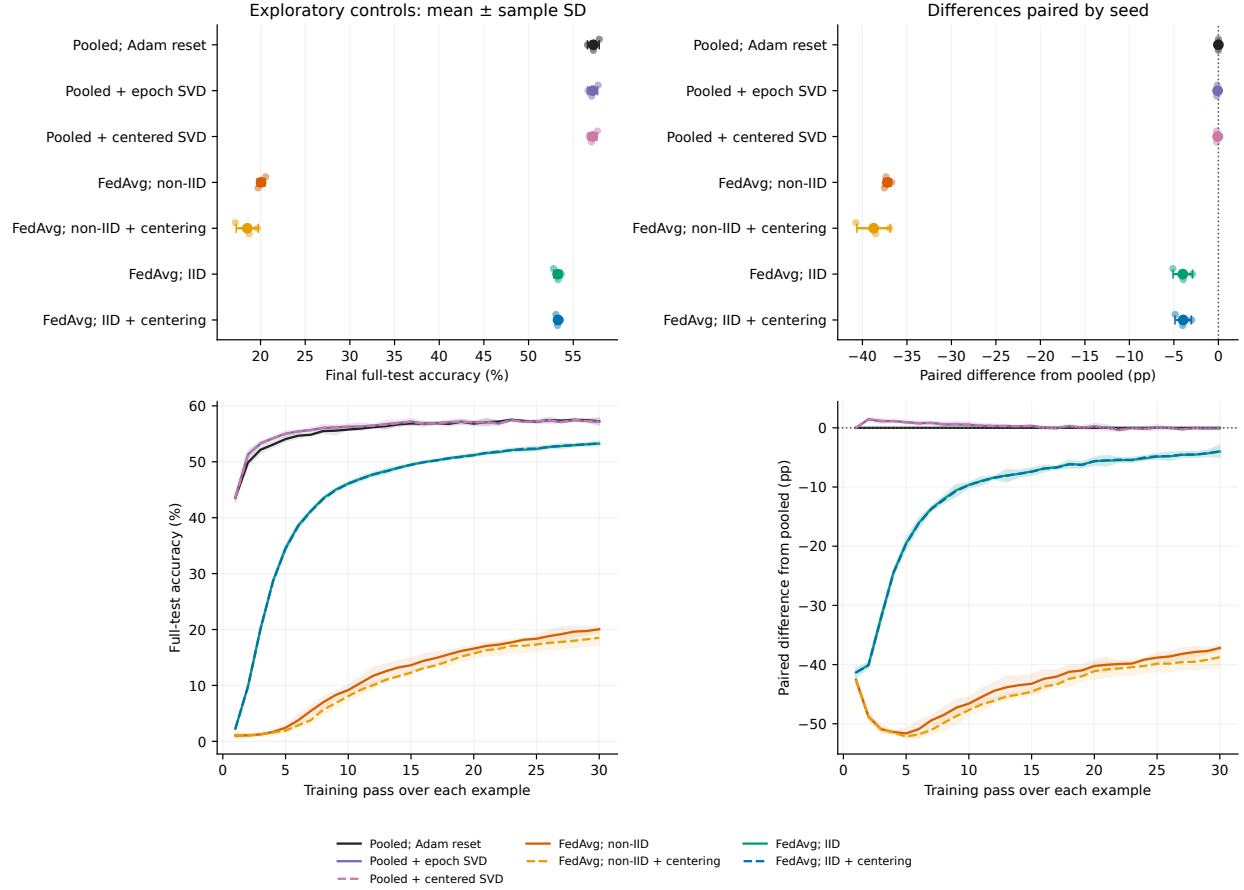


Figure 10: Paired exploratory controls for exact factor refactorization, class centering, and training-label allocation. All endpoints use the original official test set; the IID intervention preserves dataset size and every client’s sample count. These results diagnose the completed negative experiment and are not fresh confirmatory tests.

with $44.07 \pm 0.70\%$ for effective-update oracle mixing. Error feedback gives $26.63 \pm 1.68\%$ for MH and $41.28 \pm 0.71\%$ for the oracle. These are finite-run ablations of representation and compression, not universal performance orderings.

The separate online-discovery experiment selects five groups with adjusted Rand index 1.00 at stage 10 for each seed. At stage 20 its scores are 1.00, 0.918, and 1.00, with the middle seed selecting six groups. This demonstrates recovery under coordinator-visible observations; it neither establishes a privacy property nor substitutes for the fixed-ring pooled comparison.

The original Project 1 controller stayed at rank 2 throughout its five-round battery and reduced the reported training-FLOP estimate by 93.75% against fixed rank 32, while losing accuracy on three of five tasks. Since rank 32 violates every tested client ceiling, that comparison does not measure a feasible hardware-matched accuracy trade-off. The revised Fashion-MNIST rerun finishes at 82.85% for both the adaptive controller and feasible fixed ranks (4, 8, 16), with a reported 10.0% reduction in the training-FLOP estimate. This is one task, one seed, and five rounds; its unscaled adapter convention also differs from the corrected scaled-head benchmark.

Project 2’s conservative follow-up changes Spearman association with recorded contributions from 0.152 to 0.196 and pairwise ordering accuracy from 0.520 to 0.547. These are model-free

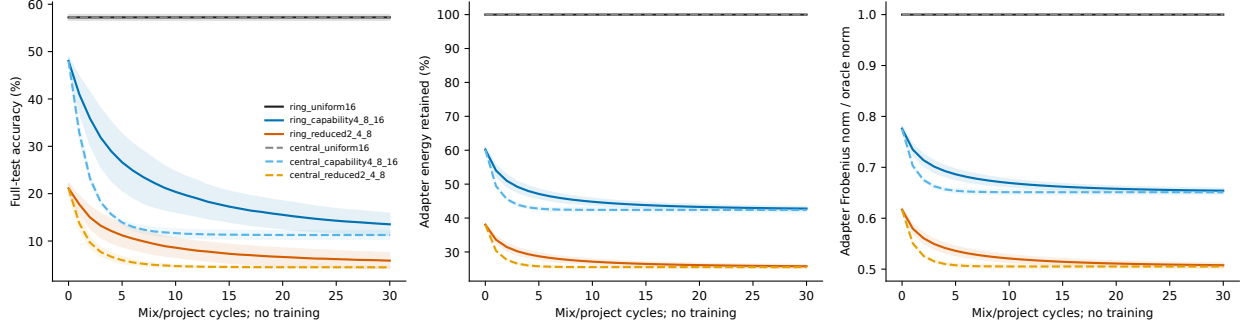


Figure 11: No-training stress test from the same completed pooled adapter at each seed. Cycle zero includes initial capacity projection and assembly. Subsequent cycles change only communication and rank projection. The uniform-rank control retains the learned classifier; repeated heterogeneous projection attenuates modes that some peers cannot store. Error bands are sample SD over three seeds.

Table 9: Supporting historical Project 3 results after 50 rounds, in percent (mean $\pm$ sample standard deviation, seeds 42–44). The objective is client-uniform; historical consensus is an evaluator-assembled uniform average. These values are not the corrected pooled comparison.

Rank context	Method	Personalized	Diagnostic consensus
Uniform 16	Local	57.93 $\pm$ 0.55	14.23 $\pm$ 1.27
	FedAvg	23.20 $\pm$ 0.29	23.20 $\pm$ 0.29
	MH	42.30 $\pm$ 1.39	15.14 $\pm$ 5.10
	Oracle hierarchy	68.98 $\pm$ 0.95	10.25 $\pm$ 0.62
Heterogeneous 4/12/32	Local	53.60 $\pm$ 0.62	14.50 $\pm$ 0.80
	FedAvg	5.74 $\pm$ 0.71	5.73 $\pm$ 0.59
	MH	28.45 $\pm$ 2.42	6.83 $\pm$ 0.45
	Oracle hierarchy	44.07 $\pm$ 0.70	11.47 $\pm$ 1.13

ranking results, not retrained-model accuracy results. Its earlier ridge forms have weak global fit ($R^2 = 0.079$ and 0.025) and task-dependent correlations. Neither the isolated Fashion-MNIST parity nor the contribution-ranking change predicts the composed method’s full-test accuracy without the corrected experiment.

7 Discussion and Limitations

The quantity-skew experiment changes the empirical picture without reversing the original negative result. Backbone LoRA on a shared task attains validation accuracy near its reconstructed decentralized references under heterogeneous adapter ranks; the previous severe collapse is therefore not an inevitable property of every decentralized LoRA implementation. Yet adaptive/sample does not beat either Dec-LoRA baseline, its final-round deficit to rank 16 is visible in the paired interval, and its traffic exceeds the feasible rank-4 baseline. This is a completed comparative study with a qualified tradeoff, not a successful demonstration of all proposed benefits.

The new factorial also gives no evidence that adaptation or size weighting improves accuracy at the chosen operating point. The mechanism control with unrestricted product-space rank performs well, but neither its rank feasibility nor its combined change in weighting and aggregation permits a

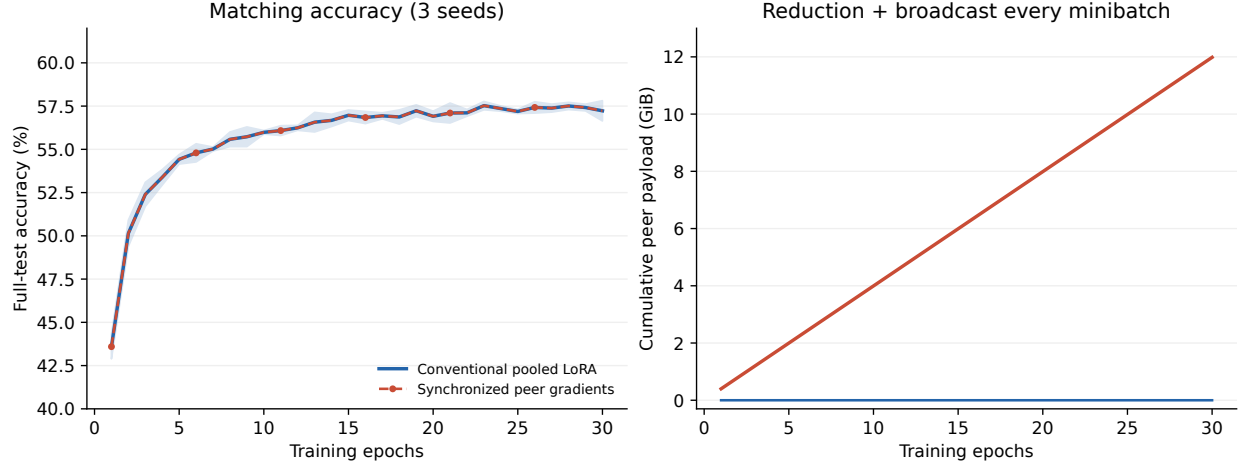


Figure 12: Uniform-rank synchronized peer factor gradients reproduce the pooled training path. Full-test predictions agree at every recorded epoch for all three seeds, with small floating-point factor differences. The control uses common Adam state and a tree collective at every globally scheduled minibatch; its payload and capacity assumptions differ from adaptive one-hop gossip.

single-component explanation. The studied sample weights are static and target conventional sample-weighted empirical risk; they do not solve dynamic domain allocation. No pooled RoBERTa/SST-2 arm was included, so even a stronger decentralized comparison would not establish the original pooled-parity claim. The one-seed equal-size anchor and previously observed CIFAR diagnostics are insufficient to attribute the new task’s behavior solely to removing label skew: the architecture, task, optimizer settings, rank scaling, and aggregation implementation also changed.

The original parity question has several distinct comparisons. Pooled versus FedAvg tests the effect of local optimization and aggregation with uniform rank. MH-16 adds neighbor communication. MH-sample imposes heterogeneous capacity. The four quality/domain arms then isolate the selected adaptive policy and the domain multipliers. A shortfall at one step cannot automatically be attributed to a different component, and matching a resource-constrained federated control does not establish matching pooled training.

The follow-up experiments replace a generic failure diagnosis with several narrower findings. Label redistribution recovers more than 33 points of uniform-rank FedAvg accuracy, while exact pooled SVD refactorization changes its pooled control by less than one tenth of a point on average. Removing common-logit offsets does not restore non-IID accuracy. The no-training test establishes repeated heterogeneous model projection as an independent mechanism for destroying retained information. The one-step Adam test shows that aggregation after local optimization can differ substantially from optimization after gradient aggregation even with no multi-step drift or SVD. These interventions identify effects in their respective settings; their magnitudes cannot be added to explain the original full-pipeline gap.

The earlier small Frobenius diagnostics remain valid but were insufficient to identify these mechanisms. FedAvg’s mean per-round relative discarded energy is 1.39×10^{-6} , and the final adaptive-domain assembly discards a mean relative energy of 7.63×10^{-5} . Neither statistic measures all predictive information removed over earlier rounds, and the uncentered norm can be dominated by a softmax-invariant offset. The uniform-rank peer arm also reaches $42.10 \pm 1.42\%$ personalized accuracy while its full-test assembled model reaches $12.69 \pm 3.55\%$, illustrating local specialization rather than accuracy of one globally useful model.

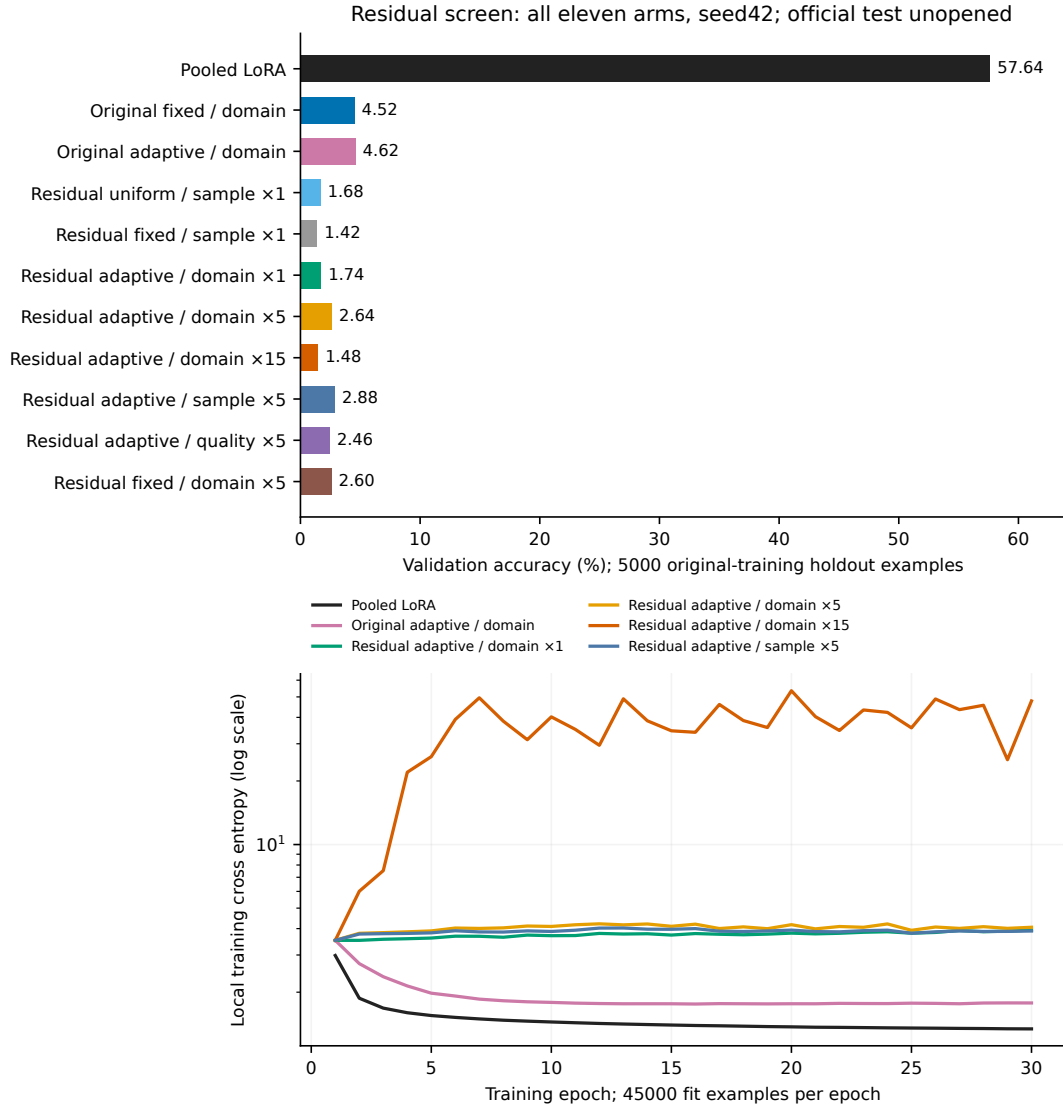


Figure 13: Completed retained-base residual screen: all eleven arms, seed 42, 30 epochs, and the common 5,000-example original-training holdout. No official-test data are opened. The upper panel retains every endpoint; the lower panel shows representative training losses, including finite instability at gain 15. Single-seed results have no between-seed error bars.

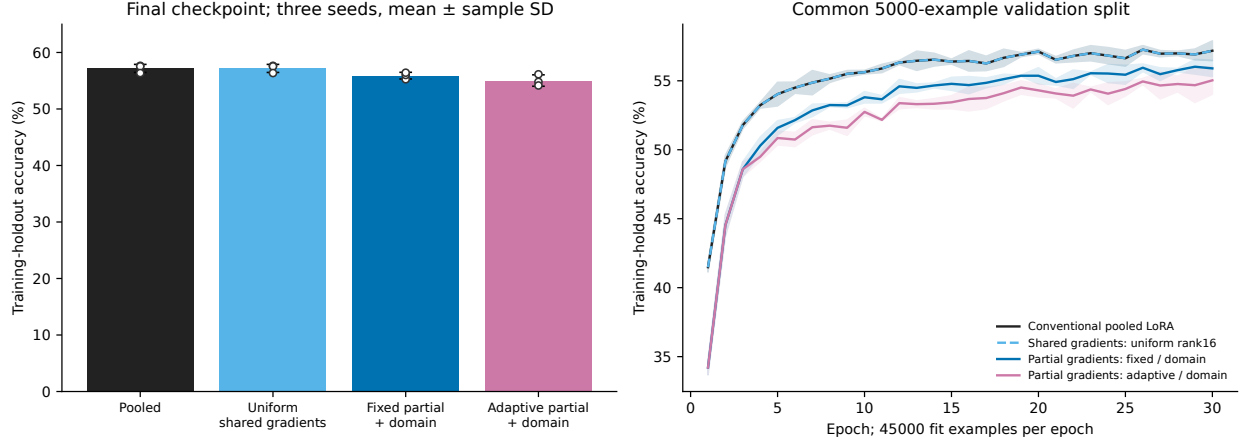


Figure 14: Shared-model partial-gradient exploration on the fixed original-training holdout. Error bars and bands are sample SD across seeds 42–44; dots show individual endpoints. The uniform driver and conventional pooled curves nearly overlap and share final prediction vectors. Fixed and adaptive partial-gradient variants retain full rank-16 model, forward and optimizer state despite their smaller gradient-coordinate budgets.

Equal global sample exposures, even with all optimizer steps counted, do not make local-then-average training equivalent to pooled sequential optimization. Local objectives differ, Adam’s gradient transformation depends nonlinearly on its moments, and SVD refactorization changes factor coordinates before the next optimizer reset. In general, if $\mathcal{U}_i$ is a local optimization map and $\mathcal{U}_{\text{pool}}$ its pooled counterpart,

$$\mathcal{C}_R \left(\sum_i \pi_i \mathcal{U}_i(\Delta W) \right) \neq \mathcal{U}_{\text{pool}}(\Delta W). \quad (49)$$

This explains why the desired equality is an empirical hypothesis rather than an algebraic property of weighted assembly. Even in a first-order SGD approximation, one local epoch with τ_i local steps gives a sample-weighted change proportional to $\sum_i (n_i/n) \tau_i \nabla f_i$, so variable τ_i changes the effective contribution of local trajectories. That approximation is not an exact description of Adam. The synchronized-gradient control establishes a sufficient route to reproduce pooled optimization, but jointly changes synchronization frequency, gradient aggregation order, and shared optimizer state. It does not identify a unique cause of FedAvg’s shortfall.

The successful uniform-rank positive control changes the scope of the conclusion: the evidence rejects the tested adaptive state-averaging protocol, not the proposition that decentralized data ownership can coexist with pooled accuracy. It nevertheless leaves the original resource claim open because every synchronized peer stores the full rank-16 model and optimizer. The completed candidate studies sharpen this distinction. All screened retained-base recipes remain far below their pooled validation reference, despite preserving a frozen accumulated model. Partial-gradient training reaches within 2.15 points on average under adaptive/domain allocation, but its fixed/domain comparator performs better, and it reduces computed gradient coordinates without reducing full model or Adam storage. Neither method satisfies the original complete-model rank cap merely because its local computation uses a smaller r_i .

Both benchmarks expose costs that a factor-only account omits. The original domain-weighted protocol exchanges dense update changes and class histograms with every peer, and its final exact

assembly uses dense matrices. The quantity protocol removes that domain-control exchange and uses compact factor merging, but its classifier traffic is substantial, root gathering stores all routed source records, and ceiling probes limit peak-memory claims. Structural workspace estimates exclude backend temporaries; measured CUDA peaks belong to a process sharing the backbone among simulated clients. Neither rank-sample products nor smaller serialized factors establish lower total FLOPs, lower independent-client peak memory, or shorter end-to-end training time.

Execution is a single-process simulation of peer message flow on the designated remote GPU machine. It verifies graph-constrained operations and payloads; it does not establish multi-host reliability, measured communication speed, asynchronous behavior, or tolerance of dropped clients. The cached-feature CIFAR-100 task is head-only, while the new study adapts internal transformer layers; neither validates medical data or arbitrary AI training. Three historical seeds and five quantity-study seeds describe limited variation, not broad equivalence. The new study’s 872 validation examples are repeatedly used to select best rounds, and its uncertainty intervals do not correct for this selection or quantify hidden-test generalization. Its fixed operating point is not a fair-tuning claim about every method’s optimum. Additional datasets, quantity ratios, topologies, and independently held-out endpoints would be needed for broader claims. The original official-test mechanism controls are exploratory follow-ups to an observed failure; the later 45,000/5,000 split limits further selection to training data but does not erase that research history. The historical oracle and automatic-discovery results have different observation assumptions and are not silently included in either fixed-ring pipeline.

Raw examples are absent from the simulated peer messages, and each client’s optimizer reads its assigned examples. The simulator process nevertheless has access to the entire dataset, so these experiments do not verify enforced isolation between machines. Updates and training statistics are exposed: the historical domain-control allgather reveals all supplied records to all peers, and the quantity-study assembly root sees every routed adapter and classifier record. No differential privacy, secure aggregation, membership-inference evaluation, reconstruction evaluation, or medical threat model is implemented. Consequently, accuracy parity, if observed, would still not justify a claim that hospitals can collaborate without privacy loss.

8 Conclusion

The completed quantity-skew comparison shows that adaptive-rank, sample-weighted decentralized LoRA can train a useful transformer: best SST-2 validation accuracy is $94.24 \pm 0.25\%$, compared with $94.54 \pm 0.24\%$ for independently reconstructed rank-16 Dec-LoRA. Its 31.81% reduction in serialized training traffic accompanies a small observed best-score deficit and a larger final-round deficit. The feasible rank-4 baseline attains $94.50 \pm 0.33\%$ with lower traffic than the proposal. Thus this experiment establishes neither a win nor statistical equivalence; the tested adaptive policy and sample weighting have not demonstrated a superior combined tradeoff. All registered runs and checkpoint verifications are complete, including unfavorable outcomes.

The earlier disjoint-label CIFAR-100 protocol remains unsuccessful at preserving pooled accuracy, with $6.90 \pm 2.03\%$ versus $57.23 \pm 0.59\%$. Its completed diagnostics identify local label allocation and repeated heterogeneous projection as important mechanisms, while exact shared-state peer gradients show that separate ownership need not itself cause accuracy loss. Residual candidates fail, and partial gradients approach pooled validation accuracy only while retaining full model and optimizer state. The new transformer benchmark asks a different comparator question and does not overturn those findings. The original joint claim of pooled accuracy under complete-model resource limits remains unsupported, and no tested protocol establishes privacy protection. These are specific

empirical limits on the evaluated methods, not an impossibility theorem for decentralized LoRA.

A Reproducibility and Supporting Figures

The completed quantity study is archived under `docs/artifacts/quantity-skew/`. The immutable source directory is `project-3-hierarchical-gossip/experiments/`. It contains the benchmark, `quantity_skew_benchmark.py`, and independent evaluator, `verify_quantity_checkpoint.py`.

The archive preserves the prospective protocol with provenance, all configurations and source manifests, pinned model/data hashes, full shard memberships, capacity and quantity permutations, per-peer sample-order hashes, per-round metrics and ranks, graph kernels, serialized-transfer byte/hash ledgers, independent checkpoint audits, and campaign status. The scientific campaign contains 36 completed full-budget runs and 72 verified checkpoint evaluations; seven separately labeled smoke runs are excluded from efficacy estimates. Large checkpoint binaries are retained remotely and identified by hashes in the repository records. The audit verifies saved best/final inference and round-metric consistency; it does not re-execute inference for every intermediate round. Final summary and figure scripts run on the designated remote host, and paper figures retain their underlying data and generation provenance.

A final source-independent artifact audit on the remote host binds all 72 saved inference audits to current checkpoint bytes, hashes the 36 additional peer-state archives, and recomputes all 720 round metric records from stored predictions and canonical validation labels. It also checks pairings, partitions, source identity, and communication arithmetic. This audit does not rerun model inference. Raw wire buffers were not retained, so saved payload hashes and ledger consistency are verified without reconstructing every historical message. Initial model-state hashes were not recorded; common initialization is supported by source-pinned same-seed code and model tests rather than a per-run initial-state artifact comparison. These limits distinguish reproducible numerical evidence from an end-to-end independent execution trace.

The historical repository records contain executable protocols, canonical policy provenance, feature and split hashes, per-round JSON records, phase timings, final adapters, and scripts that generate the reported tables and plots. The corrected driver is `project-3-hierarchical-gossip/experiments/integrated_benchmark.py`. Its evaluation does not tune on test data. Remote tests cover the numerical operators and paired execution invariants; artifacts distinguish full experiments from smoke tests and excluded pilots. Local credentials are not part of the artifact bundle.

The follow-up evidence is stored under `docs/artifacts/methodology-exploration/`, with prospective plans separating official-test diagnostics from validation-only candidate exploration. Optimization controls retain all 21 endpoints and independently reconstructed correct counts; retention runs preserve spectra, coefficient recurrence predictions, and all 18 no-training trajectories. The gradient control preserves six paired final models, 192 same-state fp64 checks, and independent predictions and communication counts. The Adam-order diagnostic retains all 24 training-only probes. Residual and partial-gradient verifiers independently reconstruct 25 unique saved validation models (11 residual-screen and 15 partial-gradient/pooled evaluations share one pooled checkpoint), with every correct count matching. They open only the original training tensor, validate its hash and exhaustive class-stratified holdout, and check exposures, initialization, rank ceilings, weight formulas and recorded transport ledgers. Per-step gradient edge traces are not retained, so their actual runtime paths cannot be independently reconstructed from the saved counters. A 47-file snapshot is verified against the six validation launch manifests and canonical P1/P2 dependency hashes. All unsuccessful candidates, the finite high-gain instability, raw per-round records and checkpoints remain archived.

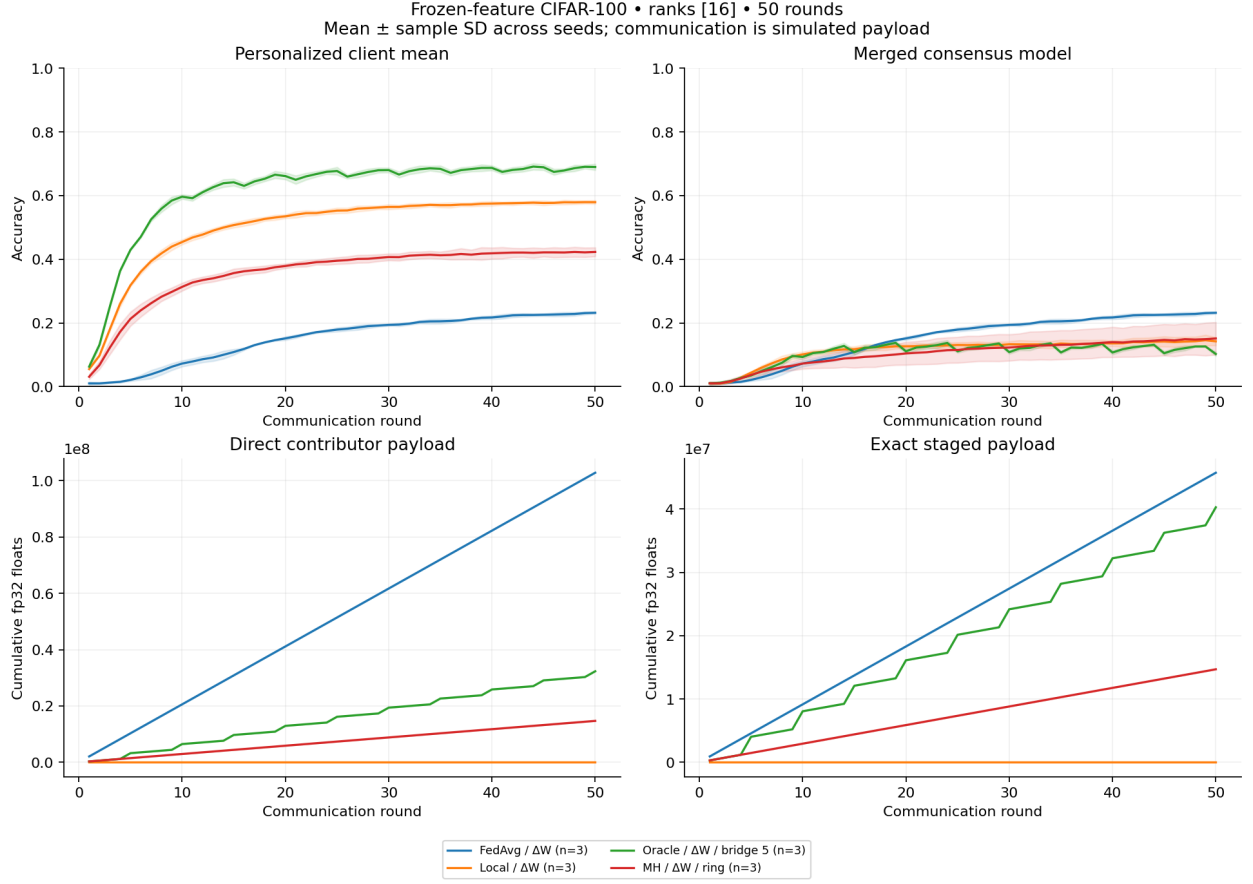


Figure 15: Supporting historical uniform-rank protocol curves, 50 rounds. Personalized and evaluator-assembled consensus accuracy are different endpoints.

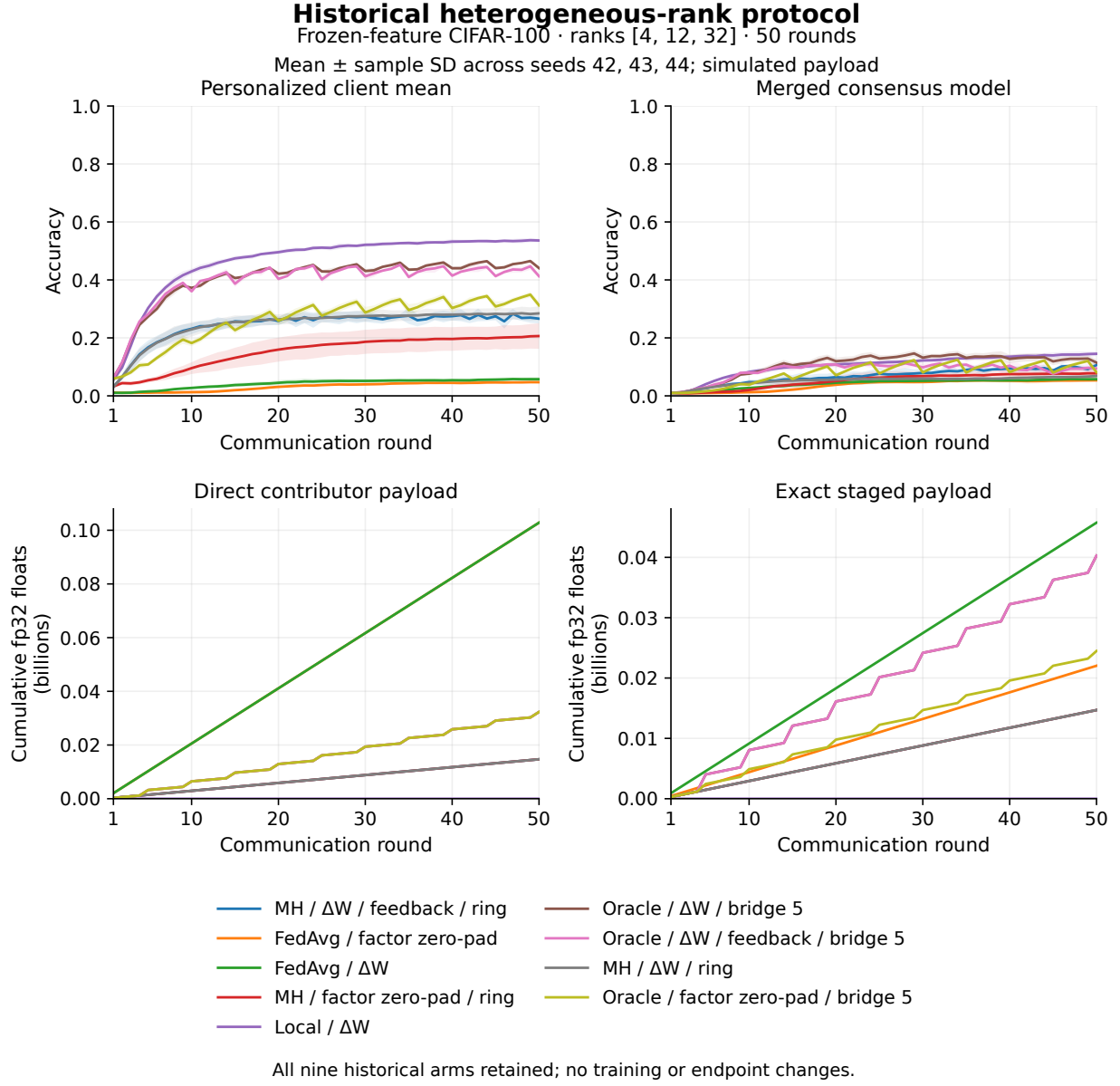
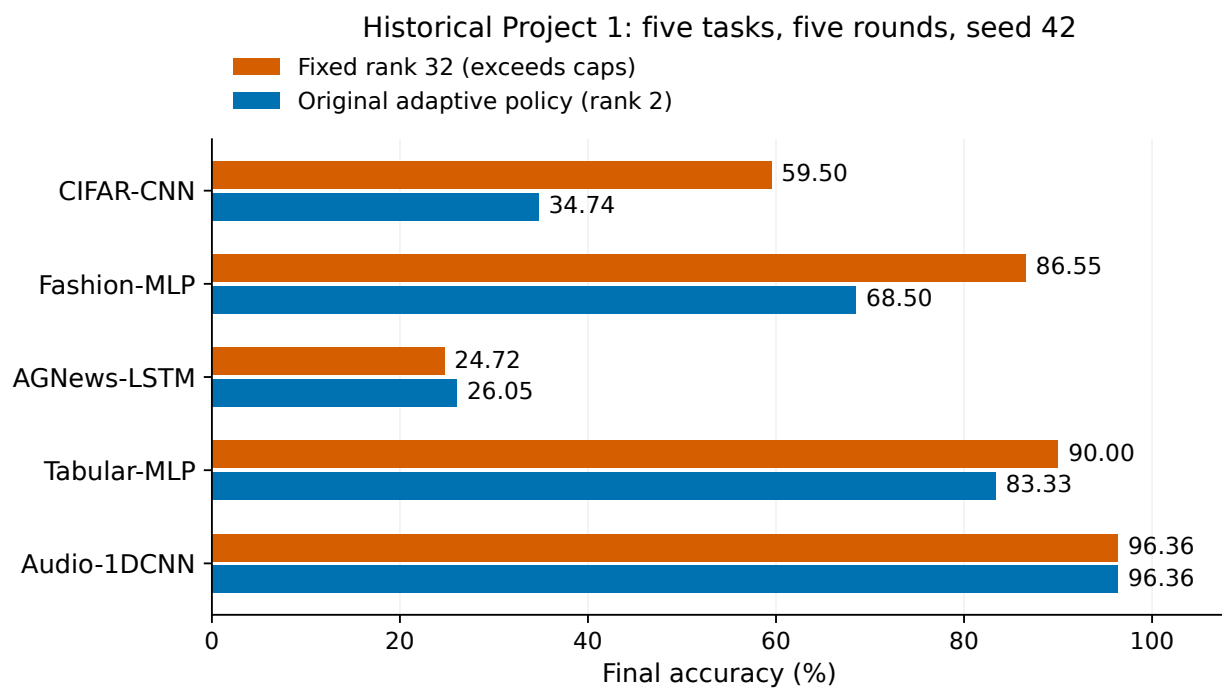


Figure 16: Supporting historical heterogeneous (4, 12, 32) protocol curves. The rank budget and objective differ from the corrected (4, 8, 16) pooled comparison.



Single-seed endpoints; no uncertainty intervals. The revised controller is reported separately.

Figure 17: Historical five-task Project 1 final accuracy after five rounds, seed 42. These single-seed endpoints have no uncertainty intervals. Fixed rank 32 exceeds the tested capability ceilings and is not a feasible hardware-matched reference. The revised Fashion-MNIST comparison is reported separately.

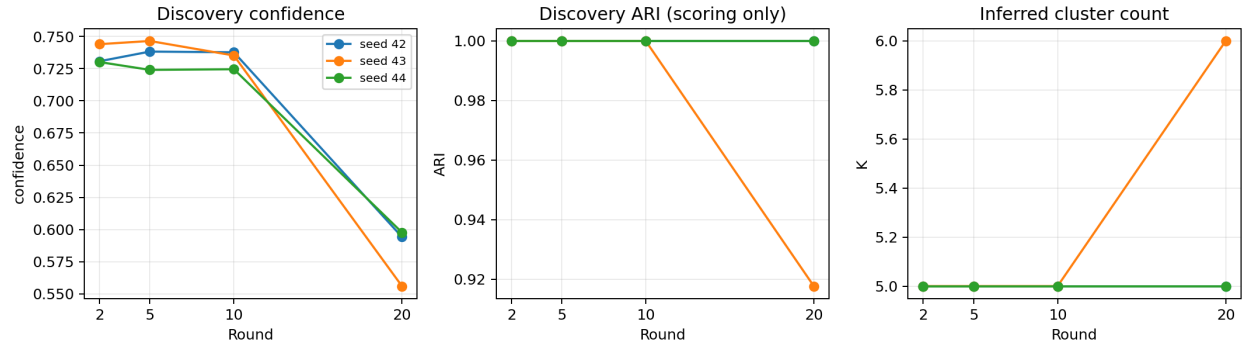


Figure 18: Supporting online-discovery results under coordinator-visible adapter observations. This experiment evaluates group recovery and is separate from the corrected fixed-ring pipeline.

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